

Systems of Linear Equations

Unit 2-1 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 2-1 slides and follow them section by section. They go further than the slides: every idea comes with extra worked examples, the traps students most often fall into are flagged in boxes titled **Watch out**, and most sections end with additional practice. Every exercise is followed by a worked solution; try each one before reading on.

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1 Why Systems of Equations?

1.1 A Problem to Motivate Us

Suppose you work in a hospital pharmacy and this request comes in:

“The pharmacy stocks a 10 mg/mL drug solution and a 2 mg/mL solution. We need 500 mL at 4 mg/mL. How much of each do we draw?”

A concentration of 10 mg/mL means that every milliliter of the solution contains 10 milligrams of the drug. Two quantities are unknown here, so we give each one a name. Let x be the number of mL of the strong solution and y the number of mL of the weak one. The problem hands us two facts that constrain them:

$$\begin{array}{ll} \text{the volumes must total 500 mL:} & x + y = 500 \\ \text{the drug must total } 4 \times 500 = 2000 \text{ mg:} & 10x + 2y = 2000 \end{array}$$

The second line deserves a word of explanation. Each mL of the strong solution carries 10 mg of drug, so x mL of it carries $10x$ mg; likewise y mL of the weak solution carries $2y$ mg. The finished 500 mL must hold 4 mg in every mL, which is 2000 mg in all.

Neither equation can be solved on its own, because each one has two unknowns in it. The first, for instance, is satisfied by $x = 100$, $y = 400$, and equally by $x = 250$, $y = 250$, and by endlessly many other pairs. Together, though, the two equations pin down exactly one pair. That pair, the x and y that satisfy **both** equations at once, is what we mean by **solving the system**.

Could we simply guess? Every pair that satisfies the first equation is a legitimate way to split the 500 mL, so we can try a few splits and work out how much drug each one delivers:

x (mL of strong)	$y = 500 - x$ (mL of weak)	drug, $10x + 2y$ (mg)	against the 2000 mg target
0	500	1000	too little
100	400	1800	too little
250	250	3000	too much
500	0	5000	too much

The right amount of strong stock is somewhere between 100 and 250 mL. We could keep narrowing the range, but guessing is slow, it never promises an exact answer, and it falls apart completely with three or four unknowns, where there is no single number to nudge up or down. What we want is a method that goes straight to the answer.

By the end of these notes we will have three ways to find it, the last of which a computer can run on a system with a thousand unknowns. We come back to this exact problem and solve it in Section 8.

1.2 Where This Shows Up in Data Science

Systems of linear equations are the workhorse underneath many of the tools you will use:

- **Fitting a model is solving a system.** Least-squares regression, the standard way to fit a line or a linear model to data, reduces to a system of linear equations in the model's coefficients.
- **Balancing and conservation.** Kirchhoff's laws for an electrical circuit, balancing a chemical reaction, and flows through a network are all linear systems. In each one some quantity is conserved, and each conservation rule is one linear equation.
- **Mixtures and dosing.** Any "how much of each" question with fixed totals, like the pharmacy problem above.
- **Redundancy detection.** A system with infinitely many solutions is telling you that your equations, or your data columns, repeat information. This is the same phenomenon as

collinearity in regression, where two predictor columns carry overlapping information and the model cannot tell their effects apart.

- **Scale.** A real problem has thousands of equations. We need a procedure a machine can follow, not cleverness.

Two of these deserve a concrete example now, since we will come back to both.

A model fit in miniature. A lab assay converts the concentration c of a substance in a sample (in mg/dL) into an instrument reading r , and over its working range the relationship is a straight line, $r = mc + b$, whose slope m and intercept b must be found for each instrument. Running two calibration standards of known concentration supplies two facts: a 10 mg/dL standard reads 0.35, and a 30 mg/dL standard reads 0.95. Substituting each pair into $r = mc + b$ gives one linear equation in the unknowns m and b :

$$\begin{cases} 10m + b = 0.35 \\ 30m + b = 0.95 \end{cases}$$

The unknowns here are called m and b , not x and y : the letters change from problem to problem, but the structure does not. Real model fitting uses many more data points and many more coefficients, yet it ends in a system of exactly this kind. We solve this one in Example 4.7.

Redundancy in a data set. Suppose a data set records each patient’s weight twice, once in kilograms and once in pounds. The pounds column is just the kilograms column multiplied by a fixed conversion factor (about 2.2), so it carries no new information. A regression that uses both columns as predictors cannot decide how to split the effect of weight between them: many different splits fit the data equally well. That “many equally good answers” is precisely a system with infinitely many solutions, and Section 7 shows how to recognize one mechanically.

Unit 1 built **functions**. A linear equation is one of the simplest there is, and this unit asks what happens when many of them must hold at once.

1.3 What We Will Build

Over the rest of these notes we build up the tools in this order:

1. What a **linear equation** is, and why its graph is a **line**.
2. The **three outcomes**: exactly one solution, none, or infinitely many.
3. **Substitution** and **elimination**: solving by hand.
4. **Three variables**: the same ideas, with planes instead of lines.
5. **Gaussian elimination**: the **augmented matrix**, the **row operations**, **row-echelon form**, then **back-substitution**.
6. Reading **no solution** or **infinitely many** straight off the rows.
7. An **application**: ticket sales at a theater.

Three notebooks go with this lecture, and the notes point to the relevant part of each as we go:

- U2-1_Systems-1_GeometryOfSolutions
- U2-1_Systems-2_SubstitutionAndElimination

2 Linear Equations and Their Graphs

2.1 What Makes an Equation Linear

Definition: Linear equation

A **linear equation** in the variables $x_1, x_2, \dots, x_n$ is one that can be written

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where the a_i (the **coefficients**) and b (the **right-hand side**) are fixed numbers.

Think of a linear equation as a recipe for a total. Each variable is an amount of something, each coefficient says how much one unit of that thing contributes, and the right-hand side is the total that all the contributions must add up to. The pharmacy equation $10x + 2y = 2000$ reads exactly this way: x mL at 10 mg per mL, plus y mL at 2 mg per mL, makes 2000 mg.

The subscripts are only labels: x_1 is “the first variable,” x_2 “the second,” and so on, which lets one definition cover any number of variables. With two or three variables we usually just call them x , y , and z .

In plain terms, every variable appears to the **first power**, alone, multiplied only by a number, and those terms are added up. That is the whole rule.

Linear	Not linear
$2x + 3y = 12$	$x^2 + y = 1$ (a power)
$x - y + 4z = 0$	$xy = 6$ (variables multiplied)
$7x = 21$	$\sqrt{x} + y = 3$ (a root)
	$\sin(x) = y$ (inside a function)

Each equation on the left fits the definition. For example, $x - y + 4z = 0$ has coefficients 1, -1 , and 4 and right-hand side 0, and $7x = 21$ is a linear equation in just one variable. Each equation on the right breaks the rule in the way noted: a variable is squared, two variables are multiplied together, or a variable sits inside a square root or a sine.

More examples, some of them in disguise. Whether an equation is linear is decided after it has been tidied up, not by how it first looks. Each of these is linear:

- $3x = 2y + 5$. Variables on both sides are fine: subtract $2y$ from both sides to get $3x - 2y = 5$, with coefficients 3 and -2 and right-hand side 5.
- $\frac{1}{2}x + 0.25y = 4$. Coefficients may be fractions or decimals; they are still fixed numbers.
- $\sqrt{2}x - \pi y = 1$. The square root is taken of the number 2, not of a variable, so $\sqrt{2} \approx 1.414$ is simply a coefficient. The same goes for $\pi \approx 3.14$.
- $2(x + 3y) = 4 - z$. Multiply out the parentheses and move z to the left: $2x + 6y + z = 4$.
- $y = 5$. A single variable is enough. In a problem whose variables are x and y , this is $0x + 1y = 5$.

- $250g + 500b = 2000$, where g counts 250 mL glasses of water and b counts 500 mL bottles that a patient drinks to meet a 2000 mL daily fluid target. Real quantities, same shape: amount times contribution, added up.

Each of these is not:

- $\frac{1}{x} + y = 3$. A variable in a denominator: $\frac{1}{x}$ is x^{-1} , a power other than the first.
- $x(y + 1) = 4$. It looks harmless, but multiplied out it is $xy + x = 4$, and xy is a product of two variables.
- $e^x + y = 2$ and $|x| + y = 1$. A variable inside a function (the exponential, the absolute value).
- Body mass index, $\text{BMI} = w/h^2$, with weight w in kilograms and height h in meters. The pairs (h, w) with a BMI of exactly 25 satisfy $w = 25h^2$: height is squared, so the set of such pairs is a curve (a parabola), not a line.

Watch out: judge the variables, and judge them after simplifying

- **Numbers can be anything; variables must stay plain.** $\sqrt{2}x$ is linear but $\sqrt{x}$ is not; $2^3x = 8x$ is linear but x^3 is not; $\sin(30^\circ)x = 0.5x$ is linear but $\sin(x)$ is not.
- **Simplify before deciding.** $x^2 - (x - 1)^2 = y$ looks nonlinear, with two squares in it. Expand: $x^2 - (x^2 - 2x + 1) = 2x - 1$, so the equation is $2x - 1 = y$, that is, $2x - y = 1$. The squares cancel and it is linear, which is why the definition says “can be written.” The trap runs the other way too: $x(y + 1) = 4$ looks tidy but hides a product.

This restriction is what makes everything in the unit work: linear problems are the ones a fixed procedure can always solve.

It is also where the “linear” in **linear regression** comes from. A linear regression model predicts an outcome as $b_0 + b_1x_1 + b_2x_2 + \cdots$: each input times a fixed number, added up. That is the left side of a linear equation.

Exercise 2.1

Which of these equations are linear? Rewrite each linear one in the form $a_1x_1 + a_2x_2 + \cdots = b$ (with the variables it uses), and for each nonlinear one say what breaks the rule.

1. $4x - 7 = 2y$
2. $x^2 + 2x = y$
3. $\frac{x}{3} - \frac{y}{5} = 1$
4. $3xy - z = 0$
5. $(x + 1)^2 - x^2 = 4y$
6. $5 = \sqrt{7}z - x$

Solution to Exercise 2.1

1. **Linear.** Subtract $2y$ from both sides and add 7: $4x - 2y = 7$.
2. **Not linear.** The x^2 term is a power other than the first.

3. **Linear.** Dividing by 3 or by 5 is multiplying by a fixed number, so the coefficients are $\frac{1}{3}$ and $-\frac{1}{5}$ and the right-hand side is 1. (Multiplying through by 15 gives the equivalent $5x - 3y = 15$.)
4. **Not linear.** xy is a product of two variables. The z term on its own would be fine, but one bad term is enough.
5. **Linear**, in disguise. $(x + 1)^2 - x^2 = (x^2 + 2x + 1) - x^2 = 2x + 1$, so the equation is $2x + 1 = 4y$, that is, $2x - 4y = -1$.
6. **Linear.** $\sqrt{7}$ is a number. Moving terms around: $-x + \sqrt{7}z = 5$.

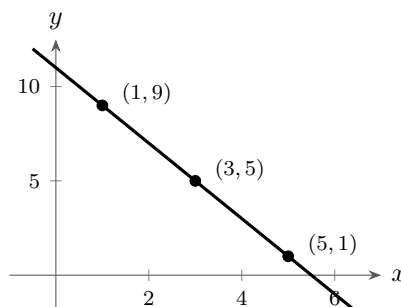
Items 5 and 6 are the Watch out above in action: simplify first, then look only at what happens to the variables.

2.2 A Linear Equation in Two Variables Is a Line

Take the equation $2x + y = 11$. Subtracting $2x$ from both sides solves it for y and puts it in the slope-intercept form $y = mx + b$ from Unit 1:

$$y = 11 - 2x.$$

So its graph is a line, with slope -2 and y -intercept 11.



The line $2x + y = 11$, with three of its infinitely many solutions.

Every point on that line is a pair (x, y) satisfying the equation, and every pair that satisfies the equation lies on the line. You can confirm any of the marked points: $(3, 5)$ works because $2(3) + 5 = 11$. The line **is** the solution set: not one answer, but infinitely many.

A single equation in two unknowns is therefore **underdetermined**: it does not carry enough information to single out one answer. To pin down one pair, we need a second equation. The notebook U2-1_Systems-1_GeometryOfSolutions draws a linear equation as a line.

Why a line, and not some other curve? In $y = 11 - 2x$, every step of 1 to the right changes y by exactly -2 , wherever you start: from $(1, 9)$ to $(2, 7)$ to $(3, 5)$. A **constant rate of change** is what makes a graph straight. In a nonlinear equation such as $y = x^2$ the change depends on where you are (from $x = 0$ to $x = 1$, y rises by 1; from $x = 2$ to $x = 3$, it rises by 5), so the graph bends.

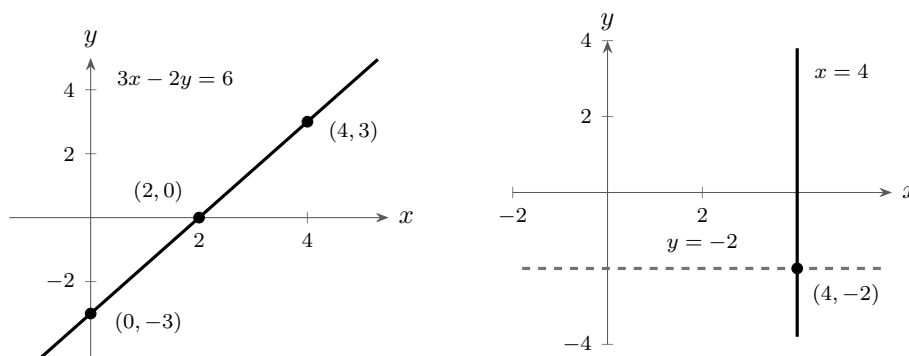
Graphing one quickly. Two points determine a line, so to draw a linear equation you need only two of its solutions. The easiest two are usually the **intercepts**, where the line crosses the axes: set $x = 0$ and solve for y , then set $y = 0$ and solve for x . A third point makes a good check.

Example 2.1 (Graphing $3x - 2y = 6$ from its intercepts). Set $x = 0$: $-2y = 6$, so $y = -3$, and the line crosses the y -axis at $(0, -3)$. Set $y = 0$: $3x = 6$, so $x = 2$, and it crosses the x -axis at $(2, 0)$. For a check, take $x = 4$: $12 - 2y = 6$ gives $-2y = -6$, so $y = 3$, and $(4, 3)$ should lie on the same straight line through the first two (it does, in the left panel below). To read off the slope, solve for y :

$$-2y = 6 - 3x \implies y = \frac{3}{2}x - 3.$$

The slope is $\frac{3}{2}$: each step of 1 to the right raises the line by 1.5. Every pair on this line, and no other pair, satisfies $3x - 2y = 6$.

Example 2.2 (Horizontal and vertical lines). The equation $y = -2$ is $0x + 1y = -2$: every point with y -coordinate -2 satisfies it, whatever x is, so its graph is a horizontal line with slope 0. The equation $x = 4$ is $1x + 0y = 4$: every point with x -coordinate 4, such as $(4, -1)$, $(4, 0)$, and $(4, 7)$, satisfies it, so its graph is a vertical line. You cannot solve $x = 4$ for y , because y does not appear in it; a vertical line has no slope-intercept form at all. The two lines in the right panel cross at $(4, -2)$, the one point satisfying both equations.



Left: $3x - 2y = 6$ drawn through its two intercepts, with $(4, 3)$ as a check. Right: the horizontal line $y = -2$ and the vertical line $x = 4$.

Watch out: not every line is $y = mx + b$

A vertical line such as $x = 4$ has no slope and cannot be written as $y = mx + b$. The form $a_1x + a_2y = b$ covers every line, vertical ones included, which is one reason these notes write linear equations that way.

Example 2.3 (A daily lab budget). A clinic can spend \$600 a day on two tests: a basic metabolic panel (BMP) at \$30 and a lipid panel at \$40. If b BMPs and ℓ lipid panels use up the budget exactly, then

$$30b + 40\ell = 600.$$

The intercepts: with no lipid panels, $30b = 600$ gives $b = 20$; with no BMPs, $40\ell = 600$ gives $\ell = 15$. Solving for ℓ gives $\ell = 15 - \frac{3}{4}b$, so the slope is $-\frac{3}{4}$: every extra BMP displaces three quarters of a lipid panel, because \$30 is three quarters of \$40. Mathematically the line has infinitely many points, but only a few make sense here, since tests come in whole, nonnegative numbers:

$$(b, \ell) = (0, 15), (4, 12), (8, 9), (12, 6), (16, 3), (20, 0).$$

Checking one: $30(8) + 40(9) = 240 + 360 = 600$. The equation describes every way to spend exactly \$600; the context then decides which of those ways are real options.

Exercise 2.2

Which of the points $(3, 0)$, $(0, -4)$, $(6, 4)$, and $(1, -3)$ lie on the line $4x - 3y = 12$? Then write the equation in slope-intercept form and state its slope.

Solution to Exercise 2.2

Substitute each point into the left side and compare with 12:

Point	$4x - 3y$	On the line?
$(3, 0)$	$12 - 0 = 12$	yes (the x -intercept)
$(0, -4)$	$0 + 12 = 12$	yes (the y -intercept)
$(6, 4)$	$24 - 12 = 12$	yes
$(1, -3)$	$4 + 9 = 13$	no

The point $(1, -3)$ misses by 1: close to the line, but not on it, and “close” does not count. Solving for y : $-3y = 12 - 4x$, so $y = \frac{4}{3}x - 4$. The slope is $\frac{4}{3}$: each step of 3 to the right raises the line by 4, which is exactly the move from $(3, 0)$ to $(6, 4)$.

2.3 A System Is an Intersection

Now add a second equation:

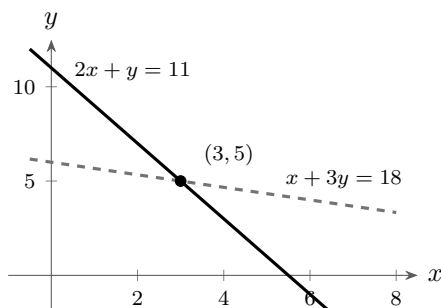
$$\begin{cases} 2x + y = 11 \\ x + 3y = 18 \end{cases}$$

The brace groups the equations into a **system**: a collection of equations that must all hold at the same time. A **solution of the system** is a pair that satisfies **both** equations, so it must lie on **both** lines. That gives the geometric picture: **solving a two-variable system means finding where the lines intersect**.

Definition: System of linear equations, and its solutions

A **system of linear equations** is a collection of linear equations in the same variables, all required to hold at once. A **solution** of the system is a choice of a value for every variable that makes **every** equation true simultaneously. The **solution set** is the collection of all such choices.

For the system above, $(3, 5)$ is a solution, because it satisfies both equations (checked below). The pair $(1, 9)$ is **not**: it satisfies the first equation, $2(1) + 9 = 11$, but in the second it gives $1 + 3(9) = 28$, not 18. Satisfying one equation is not enough; a solution must survive all of them.



Two lines, one crossing point.

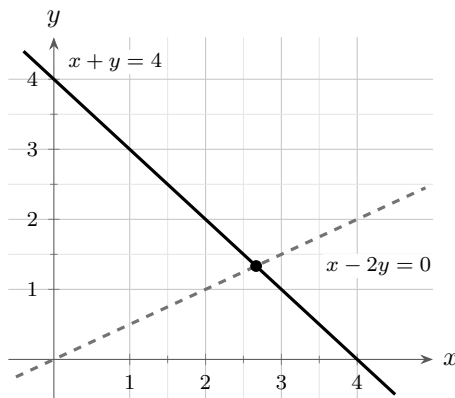
Here the lines meet at $(3, 5)$, one of the three points we marked on the first line. Of the infinitely many points on $2x + y = 11$, it is the only one that also lies on $x + 3y = 18$. Confirm it by substituting into both equations:

$$2(3) + 5 = 11 \quad \text{and} \quad 3 + 3(5) = 18.$$

Always check a solution this way. It costs ten seconds, and it catches arithmetic slips.

Example 2.4 (Testing candidate points). Is $(6, 4)$ a solution of the same system? The first equation gives $2(6) + 4 = 16 \neq 11$, so no, and we can stop there: one failed equation is enough to rule a point out. (It does satisfy the second equation, $6 + 3(4) = 18$, so it lies on the dashed line but not on the solid one.) So $(1, 9)$ sits on the solid line only, $(6, 4)$ on the dashed line only, and only the crossing point $(3, 5)$ sits on both.

Example 2.5 (Why we will not solve systems by drawing). Graph the system $x + y = 4$, $x - 2y = 0$. In slope-intercept form these are $y = 4 - x$ and $y = \frac{1}{2}x$, and the lines cross somewhere near $(2.7, 1.3)$, as the grid below suggests. Test that estimate: $2.7 + 1.3 = 4$ works, but $2.7 - 2(1.3) = 0.1$, not 0. The eye was close, but not exact. The true crossing point is $(\frac{8}{3}, \frac{4}{3}) \approx (2.667, 1.333)$, and it checks exactly: $\frac{8}{3} + \frac{4}{3} = \frac{12}{3} = 4$ and $\frac{8}{3} - 2 \cdot \frac{4}{3} = 0$. (Example 4.3 finds it by algebra.) A graph is excellent for seeing **how many** solutions there are and roughly where; for exact values we need algebra.



The crossing point is near $(2.7, 1.3)$, but a grid cannot tell you that it is exactly $(\frac{8}{3}, \frac{4}{3})$.

Exercise 2.3

Is $(2, -1)$ a solution of the system $3x + 4y = 2$, $x - y = 3$? Is $(-2, 2)$? Say what each answer tells you about where the point sits relative to the two lines.

Solution to Exercise 2.3

For $(2, -1)$: $3(2) + 4(-1) = 6 - 4 = 2$ and $2 - (-1) = 3$. Both equations hold, so **yes**: $(2, -1)$ lies on both lines, at their crossing point.

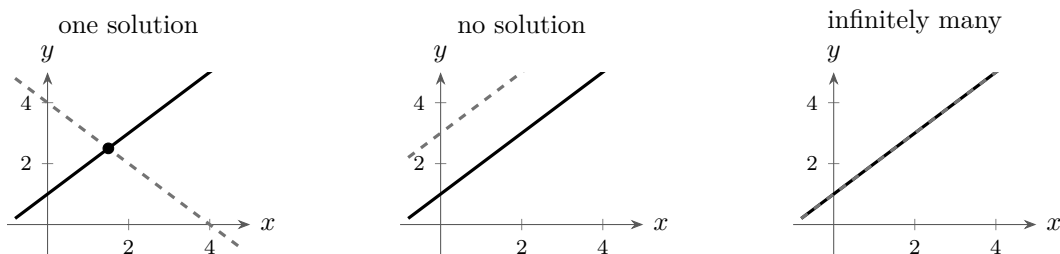
For $(-2, 2)$: $3(-2) + 4(2) = -6 + 8 = 2$, so the first equation holds, but $-2 - 2 = -4 \neq 3$, so

the second fails. **No:** $(-2, 2)$ lies on the first line only. Two different lines cross at most once, so once $(2, -1)$ is known to be the crossing point, no other point could have passed both tests.

3 The Three Outcomes

3.1 Two Lines Can Do Only Three Things

Two lines in the plane can be arranged in only three ways:



The trichotomy

1. **They cross once.** Exactly one solution. The system is **consistent** and **independent**.
2. **They are parallel and distinct.** No point lies on both, so there is **no solution**: the system is **inconsistent**.
3. **They are the same line.** Every point on it works, so there are **infinitely many** solutions. The second equation repeated the first: it is **dependent**.

A **trichotomy** is a split into exactly three cases, and the vocabulary is worth pinning down. A system is **consistent** if it has at least one solution and **inconsistent** if it has none. Its equations are **independent** when each one contributes new information, and **dependent** when one of them only repeats what the others already say. So the third case is still consistent (it has solutions, lots of them); what goes wrong there is that the second equation added nothing new.

Picture	Solutions	Consistent?	The equations
cross once	exactly one	yes	are independent
parallel and distinct	none	no	contradict each other
the same line	infinitely many	yes	are dependent (one repeats the other)

Each case has a meaning when the equations are facts about the world. **One solution:** the facts pin the unknowns down, as in the pharmacy problem. **No solution:** the facts contradict each other, so at least one of them is wrong; in a data set, that often points to an entry error, or to two measurements taken under different conditions. **Infinitely many:** the facts are redundant, so more information is needed before the unknowns can be determined.

There is no fourth case: two distinct lines cannot cross twice. If two lines shared two points they would be the same line, because two points determine a line. In particular a system of linear

equations can never have exactly two solutions, or exactly five: as soon as it has two, it has infinitely many.

The notebook `U2-1_Systems-1_GeometryOfSolutions` plots all three cases.

3.2 Telling Them Apart Before You Solve

You can tell which case you are in before doing any solving, by reading the slopes. Put both equations in the form $y = mx + b$, where m is the slope (how steep the line is) and b is the y -intercept (where it crosses the y -axis). Then:

- **Different slopes:** the lines must cross somewhere, so exactly one solution.
- **Same slope, different intercept:** parallel, never meeting, so no solution.
- **Same slope, same intercept:** one line drawn twice, so infinitely many.

Equivalently, and this is the version that survives into three or more variables, where there is no single “slope” to read: ask whether one equation is a **multiple** of the other.

- $2x + 4y = 8$ and $x + 2y = 4$: the first is exactly 2 times the second (double every term of $x + 2y = 4$ and you get $2x + 4y = 8$). The two equations say the same thing, so they describe the **same line**, and there are infinitely many solutions.
- $2x + 4y = 8$ and $x + 2y = 9$: the left sides match up to a factor of 2, but the right sides do not, since $2 \times 9 = 18$, not 8. The lines have the same tilt but sit in different places: they are **parallel**, and there is no solution.

Notice how little separates those two cases: only the right-hand side. Gaussian elimination will make the difference show up as one telltale row (Section 7).

Here are five more classifications, each showing a different wrinkle.

Example 3.1 (Reading slopes). Classify $y = 3x - 2$, $y = -x + 6$. Both are already in slope-intercept form, with slopes 3 and -1 . The slopes differ, so the lines cross exactly once: one solution. (Setting $3x - 2 = -x + 6$ gives $4x = 8$, so $x = 2$ and $y = 4$; the crossing point $(2, 4)$ checks in both, since $3(2) - 2 = 4$ and $-2 + 6 = 4$.)

Example 3.2 (Rearrange first, then compare). Classify $4x - 2y = 6$, $y = 2x - 3$. The two are written in different forms, so bring them to a common one. Solving the first for y : $-2y = 6 - 4x$, so $y = 2x - 3$, which is exactly the second equation. Same slope, same intercept: one line drawn twice, so infinitely many solutions. The multiple test agrees once the second equation is written as $2x - y = 3$: doubling it gives $4x - 2y = 6$.

Example 3.3 (The multiplier need not be a whole number). Classify $4x + 6y = 10$, $6x + 9y = 12$. Is the second a multiple of the first? Compare the matching numbers of the two equations as ratios:

$$\frac{6}{4} = 1.5, \quad \frac{9}{6} = 1.5, \quad \frac{12}{10} = 1.2.$$

The left sides agree up to the factor 1.5, so the lines are parallel (both have slope $-\frac{2}{3}$). The right sides do not follow the same factor: $1.5 \times 10 = 15$, not 12. The lines are parallel and distinct, and

there is **no solution**. Had the second equation been $6x + 9y = 15$, all three ratios would be 1.5 and the lines would coincide.

Example 3.4 (Similar-looking is not the same as proportional). Classify $2x + 3y = 6$, $3x + 2y = 6$. The same numbers appear in both, but in swapped positions, and that is not a multiple: $\frac{3}{2} \neq \frac{2}{3}$. The slopes are $-\frac{2}{3}$ and $-\frac{3}{2}$, which differ, so there is exactly one solution. (It is $(\frac{6}{5}, \frac{6}{5})$: $2 \cdot \frac{6}{5} + 3 \cdot \frac{6}{5} = \frac{30}{5} = 6$, and the second equation checks the same way.)

Example 3.5 (All three outcomes from one pharmacy shelf). Return to the pharmacy, but suppose both stock bottles on the shelf hold the same concentration, 5 mg/mL, and we are asked for 500 mL at 4 mg/mL. With x and y the volumes drawn from the two bottles,

$$\begin{cases} x + y = 500 \\ 5x + 5y = 2000 \end{cases}$$

The left side of the second equation is exactly 5 times the left side of the first, but $5 \times 500 = 2500$, not 2000. Parallel lines: **no solution**. The algebra is saying something obvious once you hear it: mixing two 5 mg/mL solutions gives 5 mg/mL whatever the proportions, so 4 mg/mL is impossible. Now change the request to 5 mg/mL. The drug total becomes $5 \times 500 = 2500$ mg, the second equation becomes $5x + 5y = 2500$, exactly 5 times the first, and there are **infinitely many** solutions: any split of the 500 mL works. With the original shelf (10 and 2 mg/mL) the slopes are -1 and -5 , which differ, so there is **exactly one** solution, found in Section 8. One shelf, three requests, three outcomes.

Watch out: compare like with like

- **Line the equations up first.** Before comparing, write both with the variables on the left, in the same order, and the constant on the right. The equations $y = 2x + 1$ and $4x - 2y = -2$ do not look related, but the second rearranges to $2y = 4x + 2$, that is, $y = 2x + 1$: the same line.
- **Zero coefficients break the ratio shortcut.** For $x = 3$ and $2x = 6$, the ratio of the y -coefficients would be $\frac{0}{0}$, which means nothing. Ask the underlying question instead: is one equation a multiple of the other? Here yes ($2x = 6$ is twice $x = 3$), so they are the same vertical line. And a vertical line against a horizontal one, such as $x = 3$ and $y = -2$, always crosses exactly once, at $(3, -2)$.

Exercise 3.1

How many solutions does each system have, and why? Answer from the slopes, without solving. For the one that has a single solution, go ahead and find it.

$$1. \begin{cases} y = 2x + 1 \\ y = 2x - 3 \end{cases} \quad 2. \begin{cases} 2x + 4y = 8 \\ x + 2y = 4 \end{cases} \quad 3. \begin{cases} x + y = 2 \\ x - y = 0 \end{cases}$$

Solution to Exercise 3.1

1. **No solution.** Both equations are already in slope-intercept form. Both slopes are 2, but the intercepts are 1 and -3 : the lines are parallel and distinct, so no pair is on both. Algebraically, a point on both lines would need its y to equal both $2x + 1$ and $2x - 3$.

Setting $2x + 1 = 2x - 3$ and subtracting $2x$ from both sides gives $1 = -3$, a contradiction: the algebra's way of saying that no such point exists.

- Infinitely many.** Divide the first equation by 2 and you get the second exactly, so both describe the same line. Solving $x + 2y = 4$ for y gives $2y = 4 - x$, that is, $y = 2 - \frac{1}{2}x$. Any point on this line solves the system: $(0, 2)$, $(2, 1)$, $(4, 0)$, and so on.
- Exactly one.** In slope form these are $y = 2 - x$ (slope -1) and $y = x$ (slope 1). Different slopes, so they cross once. To find the point, add the two equations; the y terms cancel:

$$(x + y) + (x - y) = 2 + 0 \implies 2x = 2 \implies x = 1.$$

Then $x - y = 0$ gives $y = 1$. The solution is $(1, 1)$, and it checks in both equations: $1 + 1 = 2$ and $1 - 1 = 0$.

Once the variables are all moved to the left, systems 1 and 2 both have proportional left sides: one equation's x and y coefficients are a fixed multiple of the other's. Only the right-hand side decides which of the two cases you are in.

Exercise 3.2

Without solving, classify each system as having one solution, no solution, or infinitely many, and give the reason.

- $3x - 6y = 9$ and $x - 2y = 3$
- $y = -x + 4$ and $2x + 2y = 5$
- $5x + y = 2$ and $x + 5y = 2$
- $x = 3$ and $y = -2$

Solution to Exercise 3.2

- Infinitely many.** The first equation is exactly 3 times the second, right-hand side included ($3 \times 3 = 9$): the same line.
- No solution.** Rewrite the second as $2y = -2x + 5$, that is, $y = -x + 2.5$. Both lines have slope -1 , but their intercepts are 4 and 2.5: parallel and distinct. In the multiple test, doubling the first (written $x + y = 4$) gives $2x + 2y = 8$, not 5.
- Exactly one.** The slopes are -5 (from $y = 2 - 5x$) and $-\frac{1}{5}$ (from $y = \frac{2}{5} - \frac{1}{5}x$). They differ, so the lines cross once. (The crossing point is $(\frac{1}{3}, \frac{1}{3})$: $\frac{5}{3} + \frac{1}{3} = 2$, and the second equation checks the same way.)
- Exactly one.** A vertical line and a horizontal line always cross once, here at $(3, -2)$. The system is already solved.

Exercise 3.3

Consider the system $x + 3y = 5$, $2x + ky = 10$, where k is a number you get to choose.

- For which value of k are there infinitely many solutions? For which values is there exactly one? Is there any k that gives no solution?

2. Now fix the left sides as $x + 3y = 5$, $2x + 6y = m$, and let the right-hand side m vary instead. Which values of m give infinitely many solutions, and which give none?

Solution to Exercise 3.3

1. The left sides are proportional exactly when the second is 2 times the first, which needs $k = 2 \times 3 = 6$. With $k = 6$ the right sides are proportional too ($2 \times 5 = 10$), so the two equations are the same line: **infinitely many** solutions. For every other k the slopes differ ($-\frac{1}{3}$ against $-\frac{2}{k}$, or a vertical line when $k = 0$), so there is **exactly one** solution. No value of k gives no solution: the only way to get parallel lines is $k = 6$, and then the right sides agree as well. (In fact, for every $k \neq 6$ the solution is the same point, $(5, 0)$: it satisfies $5 + 0 = 5$ and $10 + k \cdot 0 = 10$ whatever k is, because $y = 0$ switches off the only term that k touches.)
2. Now the left sides are always proportional (factor 2), so the lines are always parallel or identical, and exactly one solution is impossible. If $m = 10$ the right sides follow the same factor and there are **infinitely many** solutions; for any other m there is **no solution**.

Together the two parts are the section's point in miniature: the left sides decide whether the lines are parallel, and the right-hand side then decides between "parallel" and "identical."

4 Solving by Hand: Substitution and Elimination

4.1 The Method of Substitution

The idea of **substitution**: solve one equation for one variable, **substitute** that expression into the other equation, and the second equation is left with a single unknown, which we already know how to solve.

Substitution, step by step

1. Isolate the easiest variable (coefficient 1 is ideal, since then there is nothing to divide by).
2. Substitute into the **other** equation.
3. Solve that one-variable equation.
4. Back-substitute for the first variable.
5. Check in both originals.

As a worked example, take the system from Section 2, with its equations numbered so we can refer to them:

$$\begin{cases} 2x + y = 11 & (1) \\ x + 3y = 18 & (2) \end{cases}$$

In (1), y has coefficient 1, so that is the variable to isolate:

$y = 11 - 2x$	isolate y in (1)
$x + 3(11 - 2x) = 18$	substitute into (2)
$x + 33 - 6x = 18$	multiply out the parentheses
$-5x = -15$	collect the x terms, subtract 33
$x = 3$	divide by -5
$y = 11 - 2(3) = 5$	back into (1)

Check: $2(3) + 5 = 11$ and $3 + 3(5) = 18$. The solution is $(3, 5)$, the crossing point plotted in Section 2. The notebook U2-1_Systems-2_SubstitutionAndElimination solves this system both ways.

Substitution works on any two-variable system, but the arithmetic is kindest when some variable already has coefficient 1 or is already isolated. Here are four more, from easy to harder.

Example 4.1 (A variable already isolated). Solve $y = 2x - 1$, $3x + 2y = 12$. The first equation hands us y directly, so substitute it into the second:

$3x + 2(2x - 1) = 12$	substitute for y
$3x + 4x - 2 = 12$	multiply out
$7x = 14$	collect the x terms, add 2
$x = 2$	
$y = 2(2) - 1 = 3$	back into the first equation

Check: $2(2) - 1 = 3$ and $3(2) + 2(3) = 12$. The two lines cross at $(2, 3)$.

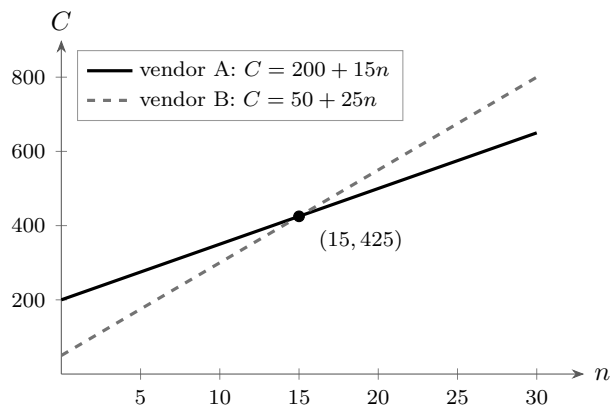
Example 4.2 (Break-even between two lab vendors). A clinic is choosing between two reference labs for a test it sends out. Vendor A charges a \$200 monthly account fee plus \$15 per test; vendor B charges a \$50 fee plus \$25 per test. For n tests in a month, the monthly cost C in dollars is

$$\begin{cases} C = 200 + 15n & (\text{vendor A}) \\ C = 50 + 25n & (\text{vendor B}) \end{cases}$$

At what volume do the two cost the same? Both equations are already solved for C , so substitute the first expression for C into the second equation; this is the same as setting the two expressions equal:

$200 + 15n = 50 + 25n$	
$150 = 10n$	subtract 50 and $15n$ from both sides
$n = 15$	
$C = 200 + 15(15) = 425$	back into vendor A's equation

Check with vendor B: $50 + 25(15) = 50 + 375 = 425$. The solution $(n, C) = (15, 425)$ is the **break-even point**: at 15 tests a month both vendors charge \$425. Below that volume vendor B's lower fee wins (at $n = 10$, A costs \$350 and B \$300); above it, vendor A's lower per-test price wins (at $n = 20$, A costs \$500 and B \$550). The crossing point of the two lines is exactly the number a manager needs.



Monthly cost against number of tests. The lines cross at the break-even point.

Example 4.3 (An answer that is not a whole number). Solve $x + y = 4$, $x - 2y = 0$, the system we could only estimate from a graph in Section 2. Adding $2y$ to both sides of the second equation isolates x : $x = 2y$. Substitute into the first:

$$\begin{aligned} 2y + y &= 4 \\ 3y &= 4 \\ y &= \frac{4}{3} \\ x = 2y &= \frac{8}{3} \end{aligned}$$

Check: $\frac{8}{3} + \frac{4}{3} = \frac{12}{3} = 4$ and $\frac{8}{3} - 2 \cdot \frac{4}{3} = 0$. The fractions are not a sign of a mistake: the exact crossing point simply does not sit at whole-number coordinates, which is exactly why reading it off a graph failed.

Example 4.4 (No coefficient equal to 1). Solve

$$\begin{cases} 3x + 2y = 4 & (1) \\ 5x + 4y = 9 & (2) \end{cases}$$

No variable has coefficient 1, so whichever one we isolate, we must divide. Isolate y in (1), since its coefficient 2 is the smallest: $2y = 4 - 3x$, so $y = 2 - \frac{3}{2}x$. Then

$$\begin{aligned} 5x + 4\left(2 - \frac{3}{2}x\right) &= 9 && \text{substitute into (2)} \\ 5x + 8 - 6x &= 9 && \text{since } 4 \cdot \frac{3}{2} = 6 \\ -x &= 1 \\ x &= -1 \\ y = 2 - \frac{3}{2}(-1) &= \frac{7}{2} && \text{back into the expression for } y \end{aligned}$$

Check: $3(-1) + 2\left(\frac{7}{2}\right) = -3 + 7 = 4$ and $5(-1) + 4\left(\frac{7}{2}\right) = -5 + 14 = 9$. The solution $\left(-1, \frac{7}{2}\right)$ has a negative coordinate and a fraction, both perfectly normal. Systems like this one, with no convenient variable to isolate, are where elimination starts to look attractive.

Watch out: substitute into the *other* equation

If you isolate $y = 11 - 2x$ from equation (1) and substitute it back into (1) itself, you get $2x + (11 - 2x) = 11$, that is, $11 = 11$. True, but useless: of course an equation agrees with a rearrangement of itself. The information about the crossing point lives in the **other** equation.

Watch out: keep the parentheses

A substituted expression goes in as a single unit, so wrap it in parentheses and multiply every term inside. Substituting $y = 11 - 2x$ into $x + 3y = 18$ gives $x + 3(11 - 2x) = x + 33 - 6x$, not $x + 33 - 2x$. Minus signs are the usual casualty: $3x - (7 - 2y)$ is $3x - 7 + 2y$, because the minus applies to both terms inside the parentheses.

When every variable disappears. Occasionally substitution eliminates *all* the variables at once. Take $y = 3x + 2$ and $6x - 2y = 5$: substituting gives $6x - 2(3x + 2) = 5$, that is, $6x - 6x - 4 = 5$, or $-4 = 5$. The x 's cancelled and what is left is false, so no pair satisfies both equations: the lines are parallel (both have slope 3). Had what was left been true, such as $5 = 5$, every point of one line would satisfy the other: the same line, and infinitely many solutions. We met this in Exercise 3.1, and Section 7 gives the systematic version.

4.2 The Method of Elimination

The idea of **elimination**: scale an equation so that a variable has **matching** coefficients in both, then subtract (or add) the equations to make that variable **vanish**.

Take the same system. Multiply (1) by 3 so that both equations contain $3y$, then subtract (2):

$$\begin{array}{rcl} 6x + 3y = 33 & 3 \times (1) & \\ - & x + 3y = 18 & (2) \\ \hline 5x & = 15 & \\ x & = 3 & \end{array}$$

Subtracting one equation from another means subtracting left side from left side and right side from right side. The two sides of each equation were equal, so the differences are equal too. Now put $x = 3$ back into (1): $2(3) + y = 11$, so $y = 5$. That is the same answer, $(3, 5)$, by a different road.

Why this is allowed. Multiplying an equation by a nonzero number, or adding one equation to another, never changes the solution set. Both moves are **reversible**: dividing by the same number undoes the first, and subtracting the added equation back off undoes the second. Since each move can be undone, nothing is gained or lost along the way: every pair that solved the old system solves the new one, and vice versa. Hold on to that fact. It is the entire justification for Gaussian elimination in Section 6.

Elimination is the method that generalizes, so it is the one we will systematize. First, three more examples, from easy to harder.

Example 4.5 (Just add). A pilot study enrolled 10 participants, with 4 more in the treatment arm than in the control arm. With t and c the sizes of the two arms,

$$\begin{cases} t + c = 10 & (1) \\ t - c = 4 & (2) \end{cases}$$

The c terms are already opposites ($+c$ and $-c$), so no scaling is needed: add the equations.

$$\begin{array}{rcl} t + c & = & 10 \quad (1) \\ + \quad t - c & = & 4 \quad (2) \\ \hline 2t & = & 14 \\ t & = & 7 \end{array}$$

Back into (1): $7 + c = 10$, so $c = 3$. Check: $7 + 3 = 10$ and $7 - 3 = 4$. Seven participants are in the treatment arm and three in the control arm.

Example 4.6 (Scale both equations). Solve

$$\begin{cases} 3x + 2y = 16 & (1) \\ 4x - 3y = 10 & (2) \end{cases}$$

Neither equation's y coefficient is a multiple of the other's (2 and -3), so scale both. The smallest number that both 2 and 3 divide into is 6, so multiply (1) by 3 and (2) by 2 to make the y terms $+6y$ and $-6y$, then add:

$$\begin{array}{rcl} 9x + 6y & = & 48 \quad 3 \times (1) \\ + \quad 8x - 6y & = & 20 \quad 2 \times (2) \\ \hline 17x & = & 68 \\ x & = & 4 \end{array}$$

Back into (1): $12 + 2y = 16$, so $y = 2$. Check: $3(4) + 2(2) = 16$ and $4(4) - 3(2) = 10$. The lines cross at $(4, 2)$. Eliminating x instead, by multiplying (1) by 4 and (2) by 3 and subtracting, gives $17y = 34$ and the same point: which variable to eliminate is a choice about arithmetic, not about the answer.

Example 4.7 (Calibrating a lab assay). Return to the calibration problem from Section 1:

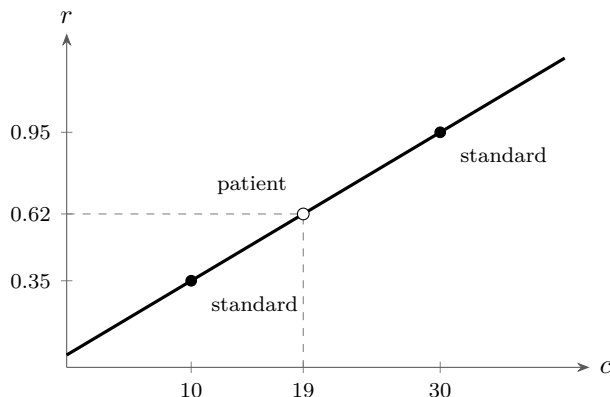
$$\begin{cases} 10m + b = 0.35 & (1) \\ 30m + b = 0.95 & (2) \end{cases}$$

Both equations contain $+b$ exactly once, so subtracting (1) from (2) removes b with no scaling at all:

$$\begin{array}{rcl} 30m + b & = & 0.95 \quad (2) \\ - \quad 10m + b & = & 0.35 \quad (1) \\ \hline 20m & = & 0.60 \\ m & = & 0.03 \end{array}$$

Back into (1): $10(0.03) + b = 0.35$, so $0.30 + b = 0.35$ and $b = 0.05$. Check in (2): $30(0.03) + 0.05 = 0.90 + 0.05 = 0.95$.

The calibration line is $r = 0.03c + 0.05$: each additional mg/dL raises the reading by 0.03, and a blank sample ($c = 0$) reads 0.05. That line is what the instrument is for. A patient sample that reads 0.62 has the concentration c for which $0.03c + 0.05 = 0.62$, so $0.03c = 0.57$ and $c = 19$ mg/dL. Two standards, one 2×2 system, and the instrument is ready to use.



The calibration line through the two standards (solid dots). A patient reading of 0.62 is traced back to a concentration of 19 mg/dL.

Watch out: every term, both sides

- **Scale every term, including the right-hand side.** Three times $2x + y = 11$ is $6x + 3y = 33$. Writing $6x + 3y = 11$ changes the equation, and the answer with it.
- **Subtract every term, signs included.** Subtracting $4x - 3y = 10$ from $4x + 5y = 26$ gives $(4x + 5y) - (4x - 3y) = 8y$ on the left, because $5y - (-3y) = 8y$; the tempting $2y$ is wrong. The right side is $26 - 10 = 16$, so $y = 2$, and then $x = 4$.

4.3 Which Method, and Why It Matters

Both methods always work on a two-variable system, but each is easier in different circumstances.

Substitution is easier when

- a variable is already isolated, or has coefficient 1;
- the system is small (two equations).

For example, take $y = 3x - 1$ and $2x + y = 9$. The first is already solved for y , so drop it into the second:

$$\begin{aligned} 2x + (3x - 1) &= 9 \\ 5x - 1 &= 9 \\ 5x &= 10 \Rightarrow x = 2, y = 3(2) - 1 = 5. \end{aligned}$$

Elimination is easier when

- every coefficient is awkward;
- there are three or more variables.

For example, take $3x + 4y = 10$ and $5x - 2y = 8$. Isolating anything makes fractions: solving the first for x , say, gives $x = \frac{10-4y}{3}$. Instead, double the second equation so that the y terms are

opposites ($+4y$ and $-4y$), and add:

$$\begin{array}{r} 3x + 4y = 10 \\ + \quad 10x - 4y = 16 \\ \hline 13x \quad = 26 \\ x \quad = 2 \end{array}$$

Then $3(2) + 4y = 10$ gives $4y = 4$, so $y = 1$.

The real reason we prefer elimination is not that it is sometimes shorter. Substitution requires you to **look** at the system and choose well; elimination is **mechanical**, with exactly one thing to do at each step. A method with no choices in it is a method a machine can run on a thousand unknowns.

Notice too that elimination never touched the letters x and y : it only scaled and added rows of **numbers**. That observation is where Gaussian elimination (Section 6) begins.

Exercise 4.1

Solve the system

$$\begin{cases} x + 2y = 7 & (1) \\ 3x - y = 7 & (2) \end{cases}$$

1. once by **substitution**,

2. once by **elimination**,

and confirm that the two agree. Then say, without plotting, what the graph looks like. Check your answer in **both** original equations before moving on.

Solution to Exercise 4.1

By substitution. In (1), x has coefficient 1, so isolate x and substitute into (2):

$x = 7 - 2y$	isolate x in (1)
$3(7 - 2y) - y = 7$	substitute into (2)
$21 - 6y - y = 7$	multiply out
$-7y = -14$	subtract 21
$y = 2$	divide by -7
$x = 7 - 2(2) = 3$	back-substitute

By elimination. Double (2) so that the y terms are opposites ($+2y$ in (1), $-2y$ in the doubled (2)), then add:

$$\begin{array}{r} x + 2y = 7 \quad (1) \\ + \quad 6x - 2y = 14 \quad 2 \times (2) \\ \hline 7x \quad = 21 \\ x \quad = 3 \end{array}$$

Back into (1): $3 + 2y = 7$, so $2y = 4$ and $y = 2$.

Answer. Both roads give $(x, y) = (3, 2)$. Check: $3 + 2(2) = 7$ and $3(3) - 2 = 7$.

For the graph, put each equation in slope-intercept form: (1) becomes $y = \frac{7}{2} - \frac{1}{2}x$, with slope $-\frac{1}{2}$, and (2) becomes $y = 3x - 7$, with slope 3. The slopes differ, so the graph is **two lines crossing once**, at $(3, 2)$.

More practice. Four more, including a word problem and one with a twist.

Exercise 4.2

Solve $y = 4 - 2x$, $5x - 3y = 10$ by substitution.

Solution to Exercise 4.2

The first equation already isolates y , so substitute it into the second, keeping the parentheses:

$$5x - 3(4 - 2x) = 10$$

$$5x - 12 + 6x = 10$$

the minus sign reaches both terms

$$11x = 22$$

$$x = 2$$

$$y = 4 - 2(2) = 0$$

Check: $0 = 4 - 4$ and $5(2) - 3(0) = 10$. The solution is $(2, 0)$. A coordinate of 0 is a real answer, not a failure: the two lines cross on the x -axis.

Exercise 4.3

Solve $4x + 3y = 5$, $6x - 5y = 17$ by elimination.

Solution to Exercise 4.3

Eliminate x . The smallest number that both 4 and 6 divide into is 12, so multiply the first equation by 3 and the second by 2, then subtract:

$$\begin{array}{rclcl} 12x + 9y & = & 15 & 3 \times \text{first} \\ - & 12x - 10y & = & 34 & 2 \times \text{second} \\ \hline & 19y & = & -19 & \\ & y & = & -1 & \end{array}$$

The y column is $9y - (-10y) = 19y$, and the right side is $15 - 34 = -19$. Back into the first equation: $4x + 3(-1) = 5$, so $4x = 8$ and $x = 2$. Check: $4(2) + 3(-1) = 5$ and $6(2) - 5(-1) = 12 + 5 = 17$. The solution is $(2, -1)$.

Exercise 4.4

A clinic ordered 40 boxes of exam gloves, some size small at \$8 a box and the rest size large at \$11 a box, and spent \$377. How many boxes of each size did it order?

Solution to Exercise 4.4

Let s and ℓ be the numbers of boxes of small and large. The two facts are a count and a cost:

$$\begin{cases} s + \ell = 40 & \text{(boxes)} \\ 8s + 11\ell = 377 & \text{(dollars)} \end{cases}$$

Multiply the count equation by 8, giving $8s + 8\ell = 320$, and subtract it from the cost equation: $3\ell = 57$, so $\ell = 19$. Then $s = 40 - 19 = 21$. Check: $21 + 19 = 40$ boxes, and $8(21) + 11(19) =$

$168 + 209 = 377$ dollars. The clinic ordered 21 boxes of small and 19 of large. (Substitution works as well: $s = 40 - \ell$ in the cost equation gives $320 - 8\ell + 11\ell = 377$, the same $3\ell = 57$.)

Exercise 4.5

Solve $y = \frac{1}{2}x + 3$, $x - 2y = -6$ by substitution. What happens, and what does it mean?

Solution to Exercise 4.5

Substitute the first into the second:

$$x - 2\left(\frac{1}{2}x + 3\right) = -6 \implies x - x - 6 = -6 \implies -6 = -6.$$

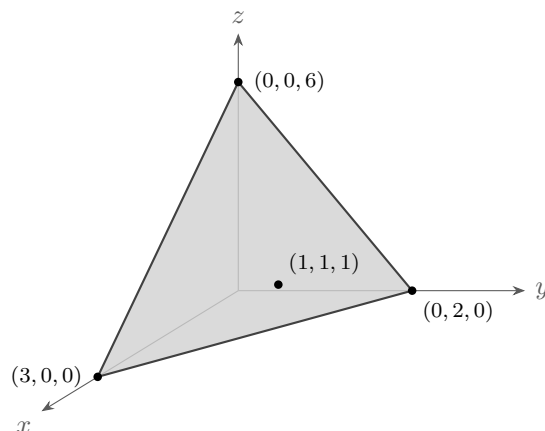
Every variable vanished and what is left is **true**, whatever x is. That means every point of the first line also satisfies the second equation: they are the same line. Indeed, solving $x - 2y = -6$ for y gives $2y = x + 6$, that is, $y = \frac{1}{2}x + 3$. There are **infinitely many** solutions, all the points $\left(x, \frac{1}{2}x + 3\right)$, such as $(0, 3)$, $(2, 4)$, and $(-6, 0)$. Compare the system $y = 3x + 2$, $6x - 2y = 5$ at the end of Section 4.1, where the leftover statement $-4 = 5$ was false and there was no solution at all.

5 Three Variables

5.1 In Three Variables, an Equation Is a Plane

An equation like $2x + 3y + z = 6$ has three unknowns, so each solution is a triple (x, y, z) : a point in three-dimensional space. The point $(1, 1, 1)$ is one, since $2 + 3 + 1 = 6$. All of its solutions together form a whole **plane**, a flat surface extending forever in every direction, just as a two-variable equation had a whole line of solutions.

Example 5.1 (Picturing the plane $2x + 3y + z = 6$). In three variables the intercepts are found the same way as in two: set two of the variables to 0 and solve for the third. With $y = z = 0$, $2x = 6$ gives $(3, 0, 0)$; with $x = z = 0$, $3y = 6$ gives $(0, 2, 0)$; with $x = y = 0$, $z = 6$ gives $(0, 0, 6)$. The triangle joining these three points is the piece of the plane where no coordinate is negative, and it contains $(1, 1, 1)$. A non-example: $(1, 1, 2)$ gives $2 + 3 + 2 = 7 \neq 6$, so it is not on the plane. Above the spot $x = 1$, $y = 1$ the plane sits at height $z = 6 - 2 - 3 = 1$, and this point is at height 2, just above it.



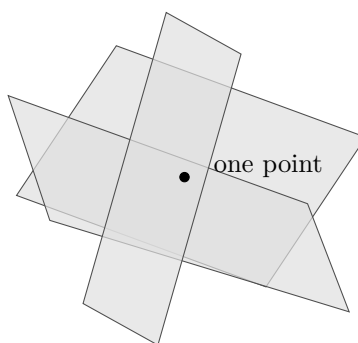
The part of the plane $2x + 3y + z = 6$ with no negative coordinates: a triangle through the three intercepts.

Example 5.2 (A daily calorie target). Carbohydrate and protein supply about 4 kcal per gram and fat about 9 kcal per gram. A day's intake of c grams of carbohydrate, p grams of protein, and f grams of fat hits a 2000 kcal target exactly when

$$4c + 4p + 9f = 2000,$$

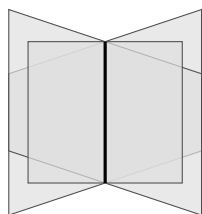
a linear equation in three variables, so its solutions form a plane. The day $(c, p, f) = (230, 90, 80)$ is on it: $920 + 360 + 720 = 2000$. The day $(250, 100, 50)$ is not: $1000 + 400 + 450 = 1850$. A single calorie target leaves a whole plane of possible diets. A dietitian who also sets a protein target and a fat target has three equations, and three planes can meet in a single point.

So a system of three equations in three unknowns asks: *where do these three planes all meet?*

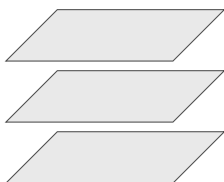


Three planes crossing at a single point: the one-solution case.

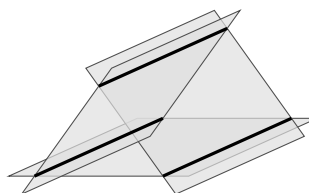
The outcomes are still three. Three planes can meet in a single **point** (one solution); in a whole **line**, like pages of an open book meeting along the spine, or even a whole **plane** (infinitely many solutions); or **nowhere** at all (no solution). The sketches below show a common line and two different ways of meeting nowhere.



a common line:
infinitely many



three parallel planes:
no solution



each pair meets, all three never:
no solution

Watch out: “no solution” does not require parallel planes

In two variables, no solution meant parallel lines. In three, the planes can fail to share a point even when no two of them are parallel. In the right-hand sketch each pair of planes meets in a line, but the three lines of intersection run side by side, like the edges of a triangular tent, and never meet. Section 7 works through a system that does exactly this (Example 7.1). A related fact: **two** planes can never meet in just one point. Two different planes are either parallel or meet in a whole line, so two equations in three unknowns can never have exactly one solution.

Drawing this by hand gets hard fast, which is another reason to want a procedure that does not depend on picturing the geometry. The notebook `U2-1_Systems-1_GeometryOfSolutions` has rotatable 3-D planes you can turn to see each case.

5.2 Solving a Three-Variable System by Elimination

Solve

$$\begin{cases} x + y + z = 6 & (1) \\ 2x - y + z = 3 & (2) \\ x + 2y - z = 2 & (3) \end{cases}$$

Strategy: use (1) to knock x out of (2) and (3). That leaves two equations in only y and z , a 2×2 system (two equations, two unknowns) of exactly the kind we just solved. Finish it, then back-substitute.

Eliminate x . Equation (2) has $2x$ and (1) has x , so subtract twice (1) from (2). Equation (3) and (1) both have x , so subtract (1) from (3). We call the results (2') and (3'), read “two prime” and “three prime,” to mark them as new versions of (2) and (3):

$$\begin{array}{rcl} 2x - y + z = 3 & (2) & \\ - & 2x + 2y + 2z = 12 & 2 \times (1) \\ \hline -3y - z = -9 & (2') & \end{array} \qquad \begin{array}{rcl} x + 2y - z = 2 & (3) & \\ - & x + y + z = 6 & (1) \\ \hline y - 2z = -4 & (3') & \end{array}$$

Eliminate y . Now (2') has $-3y$ and (3') has y . Tripling (3') gives $3y$, which cancels $-3y$ when the two are added, so compute $(2') + 3 \times (3')$:

$$\begin{array}{rcl} -3y - z = -9 & (2') & \\ + & 3y - 6z = -12 & 3 \times (3') \\ \hline -7z = -21 & & \\ z = 3 & & \end{array}$$

Back-substitute, bottom up:

$$\begin{aligned}(3') : \quad y - 2(3) &= -4 \Rightarrow y = 2 \\ (1) : \quad x + 2 + 3 &= 6 \Rightarrow x = 1\end{aligned}$$

Any equation with only y and z left in it would do for the first line; $(2')$ gives the same result, since $-3y - 3 = -9$ also means $y = 2$.

Answer: $(x, y, z) = (1, 2, 3)$. Check all three original equations: $1 + 2 + 3 = 6$; $2 - 2 + 3 = 3$; $1 + 4 - 3 = 2$.

The same strategy works on every three-variable system. Two more, from easy to harder.

Example 5.3 (A missing variable saves work). Solve

$$\begin{cases} x + y + z = 9 & (1) \\ y - z = 1 & (2) \\ 2x + y + z = 13 & (3) \end{cases}$$

Equation (2) has no x in it already, so only (3) needs its x removed. It has $2x$, so subtract twice (1):

$$\begin{array}{rcl} 2x + y + z & = & 13 \quad (3) \\ - & 2x + 2y + 2z & = 18 \quad 2 \times (1) \\ \hline & -y - z & = -5 \quad (3') \end{array}$$

Now (2) and (3') form a 2×2 system in y and z . Their y terms are opposites, so add them: $(y - z) + (-y - z) = 1 + (-5)$, that is, $-2z = -4$, so $z = 2$. Back-substitute: (2) gives $y - 2 = 1$, so $y = 3$; (1) gives $x + 3 + 2 = 9$, so $x = 4$. Check all three: $4 + 3 + 2 = 9$; $3 - 2 = 1$; $8 + 3 + 2 = 13$. The solution is $(4, 3, 2)$. (A sharp eye might spot a shortcut here: $(3) - (1)$ is simply $x = 4$. Shortcuts are welcome when you see them, but the systematic route always works, whether you see them or not.)

Example 5.4 (Every coefficient in play). Solve

$$\begin{cases} x + 2y + z = 3 & (1) \\ 2x + y - z = 0 & (2) \\ 3x - y + 2z = 13 & (3) \end{cases}$$

Eliminate x . Subtract twice (1) from (2), and three times (1) from (3):

$$\begin{aligned}(2) - 2 \times (1) : \quad (2x - 2x) + (y - 4y) + (-z - 2z) &= 0 - 6, \quad \text{that is,} \quad -3y - 3z = -6, \\ (3) - 3 \times (1) : \quad (3x - 3x) + (-y - 6y) + (2z - 3z) &= 13 - 9, \quad \text{that is,} \quad -7y - z = 4.\end{aligned}$$

Every term in the first result is divisible by -3 , so divide through to get the simpler $y + z = 2$ ($2'$). Call the second result ($3'$): $-7y - z = 4$.

Eliminate y . ($2'$) has y and ($3'$) has $-7y$, so compute ($3'$) + $7 \times (2')$:

$$(-7y - z) + (7y + 7z) = 4 + 14 \implies 6z = 18 \implies z = 3.$$

Back-substitute. ($2'$): $y + 3 = 2$, so $y = -1$. (1): $x + 2(-1) + 3 = 3$, so $x = 2$.

Check in the originals: $2 - 2 + 3 = 3$; $4 - 1 - 3 = 0$; $6 + 1 + 6 = 13$. The three planes meet at the single point $(2, -1, 3)$. Dividing ($2'$) by -3 was optional, but shrinking the numbers whenever an entire equation shares a factor makes every later step easier.

5.3 The Pattern Worth Noticing

Step back from the arithmetic and look at the order in which things happened.

Elimination works downward, then solves upward

1. Use equation 1 to remove x from equations 2 and 3.
2. Use the **new** equation 2 to remove y from the new equation 3.
3. The newest equation 3 has only z in it: solve it for z .
4. Plug z into the new equation 2 and solve for y .
5. Plug y and z into equation 1 and solve for x .

Steps 1 and 2 are the downward sweep, called **forward elimination**; steps 3 to 5 are the upward pass, called **back-substitution**. The target shape is **triangular**:

$$\begin{cases} x + y + z = 6 \\ -3y - z = -9 \\ -7z = -21 \end{cases}$$

Each equation has one fewer variable than the one above it, so the last one hands you an answer for free: $-7z = -21$ gives $z = 3$ at once. Each answer then unlocks the equation above it.

Example 5.5 (Back-substitution on its own). A system that already has the triangular shape needs no elimination at all, only the upward pass:

$$\begin{cases} x + 2y - z = 5 \\ y + 3z = 11 \\ 2z = 6 \end{cases}$$

The bottom equation gives $z = 3$. The middle one then reads $y + 9 = 11$, so $y = 2$. The top one reads $x + 4 - 3 = 5$, so $x = 4$. Check: $4 + 4 - 3 = 5$; $2 + 9 = 11$; $6 = 6$. The solution is $(4, 2, 3)$, found without a single elimination step: getting to this shape is the hard part, and forward elimination exists to get there.

Watch out: back-substitution starts at the bottom

The top equation of a triangular system has all three unknowns in it, so starting there gets you nowhere: $x + 2y - z = 5$ alone has a whole plane of solutions. Always begin with the last equation, the one with a single unknown, and work up.

Exercise 5.1

Solve by elimination, and check your answer in all three original equations:

$$\begin{cases} x + y - z = 0 & (1) \\ 2x - y + z = 6 & (2) \\ x + 3y + 2z = 11 & (3) \end{cases}$$

Solution to Exercise 5.1

Eliminate x . (2) $- 2 \times$ (1): $(-y - 2y) + (z + 2z) = 6 - 0$, that is, $-3y + 3z = 6$; divide by -3 to get $y - z = -2$ (2'). (3) $-$ (1): $(3y - y) + (2z + z) = 11 - 0$, that is, $2y + 3z = 11$ (3').

Eliminate y . (3') $- 2 \times$ (2'): $(2y - 2y) + (3z + 2z) = 11 + 4$, so $5z = 15$ and $z = 3$.

Back-substitute. (2'): $y - 3 = -2$, so $y = 1$. (1): $x + 1 - 3 = 0$, so $x = 2$.

Check: $2 + 1 - 3 = 0$; $4 - 1 + 3 = 6$; $2 + 3 + 6 = 11$. The solution is $(2, 1, 3)$. (A shortcut was available: adding (1) and (2) cancels both y and z at once, giving $3x = 6$. Worth spotting, never necessary.)

6 Gaussian Elimination

6.1 First, Strip Away the Letters: the Augmented Matrix

The nuisance in that elimination was bookkeeping: we kept rewriting x , y , z , and $=$ when only the **numbers** changed. So let us stop writing the letters.

Definition: Augmented matrix

The **augmented matrix** of a system is the grid of its coefficients, with the right-hand sides attached as a final column, separated by a bar.

$$\begin{cases} x + y = 3 \\ 2x + y = 5 \end{cases} \longleftrightarrow \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array} \right]$$

Each **row** is one equation. Each **column** left of the bar belongs to one variable (here the first holds the x coefficients and the second the y coefficients), and the column to the right of the bar holds the constants. The bar stands where the $=$ signs were. So the second row, $[2 \ 1 \mid 5]$, is just a compact way of writing $2x + 1y = 5$.

An augmented matrix is nothing but a way of storing the **coefficients** and the **right-hand side** without the clutter. In data-science terms it is a small table: one record (row) per equation, one field (column) per variable, plus a final field for the constant.

Watch out: a missing variable is a zero, not a blank

If a variable does not appear in an equation, its coefficient is 0, and that 0 must be written in:

$$\begin{cases} x & + z = 4 \\ & y & = 2 \end{cases} \longleftrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{array} \right]$$

The first equation has no y , so its middle entry is 0; the second has only y , so its x and z entries are 0. Keep the columns aligned, or the bookkeeping fails silently: a number that drifts into the wrong column becomes the coefficient of the wrong variable, and nothing will warn you.

Example 6.1 (Tidy up first, then write the matrix). Write the augmented matrix of

$$\begin{cases} 2x - y + 3z = 5 \\ y = 4 - x \\ 3z - 2 = x \end{cases}$$

The first equation is already in the shape “variables on the left, in the order x, y, z ; a number on the right.” The second has x on the wrong side: add x to both sides to get $x + y = 4$, and since there is no z , that is $x + y + 0z = 4$. The third has a constant on the left and a variable on the right: subtract x from both sides and add 2, giving $-x + 3z = 2$, which is $-x + 0y + 3z = 2$. Now read off the coefficients in the order x, y, z :

$$\begin{cases} 2x - y + 3z = 5 \\ x + y = 4 \\ -x + 3z = 2 \end{cases} \longleftrightarrow \left[\begin{array}{ccc|c} 2 & -1 & 3 & 5 \\ 1 & 1 & 0 & 4 \\ -1 & 0 & 3 & 2 \end{array} \right]$$

Example 6.2 (From the matrix back to the equations). With variables x, y, z , the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 3 & 1 & 6 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

stands for the system $x - 2y = 0$, $3y + z = 6$, $2z = 4$. A 0 to the left of the bar means that variable is absent from that equation, and a 0 to the right of the bar means the right-hand side is zero; neither one means “nothing to see here.” This particular system happens to be triangular already: $2z = 4$ gives $z = 2$, then $3y + 2 = 6$ gives $y = \frac{4}{3}$, then $x = 2y = \frac{8}{3}$.

A non-example. The system $x + y = 3$, $xy = 2$ has no augmented matrix. The columns are reserved for x and for y , and the term xy belongs to neither. Augmented matrices, and everything built on them in this section, are for linear systems only.

Watch out: one column per variable, in the same order every time

Before writing rows, put every equation’s variables in the same order. The system $x + 2y = 5$, $3y + x = 7$ has second row $[1 \ 3 \mid 7]$, because its x coefficient is 1 and its y coefficient is 3. Copying the numbers in the order they happen to be printed gives $[3 \ 1 \mid 7]$, which is the equation $3x + y = 7$: a different system, with a different answer ($(\frac{9}{5}, \frac{8}{5})$ instead of the correct $(1, 2)$), and no error message anywhere.

For now an augmented matrix is a **grid of numbers with a bar in it**, nothing more. What a matrix *is*, and what it does to a vector, waits for Unit 2-3.

Exercise 6.1

1. Write the augmented matrix of this system, using the variable order a, b, c :

$$\begin{cases} a + 2c = 7 \\ 3b - a = 4 - c \\ 2a = b + 1 \end{cases}$$

2. With variables x, y, z , write out the system whose augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 4 & -3 \end{array} \right].$$

Solution to Exercise 6.1

1. Rearrange each equation to have its variables on the left, in the order a, b, c . The first has no b : $a + 0b + 2c = 7$. For the second, add c to both sides and reorder: $-a + 3b + c = 4$. For the third, subtract b from both sides: $2a - b + 0c = 1$. So

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 7 \\ -1 & 3 & 1 & 4 \\ 2 & -1 & 0 & 1 \end{array} \right].$$

2. Row 1 is $1x + 0y - 1z = 2$, that is, $x - z = 2$. Row 2 is $0x + 1y + 4z = -3$, that is, $y + 4z = -3$. Two equations in three unknowns: as the Watch out in Section 5 noted, such a system can never pin down a single point.

6.2 Row-Echelon Form: the Target Shape

The three-variable example ended at a triangular system. In matrix language, that target shape has a name.

Definition: Row-echelon form

A matrix is in **row-echelon form** when

1. Any all-zero rows sit at the bottom.
2. In each nonzero row, the first nonzero entry (the **pivot**) is to the **right** of the pivot in the row above.
3. Every entry **below** a pivot is zero.

“Echelon” is a military word for a formation arranged in steps, and the result is exactly that: the staircase, or triangular, shape we reached by hand. Below, a $*$ marks an entry whose value does not matter for the shape:

$$\left[\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array} \right] \qquad \left[\begin{array}{ccc|c} * & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{array} \right]$$

Both are valid: the pivots must move right as you go down, but need not move by exactly one column. In the second matrix, the second row’s pivot skips column 2 entirely and sits in column 3.

Notice what the definition does **not** ask for. It does not require the pivots to be 1 (the algorithm below makes them 1 for convenience, but a pivot of 2 or -4 is still a pivot), and it places no condition at all on the entries above a pivot or to the right of the bar. Row-echelon form is only about the zeros **below** the staircase.

Example 6.3 (Which of these are in row-echelon form?).

$$\begin{array}{lll} \text{(a)} \left[\begin{array}{ccc|c} 1 & 3 & -1 & 2 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 1 & -2 \end{array} \right] & \text{(b)} \left[\begin{array}{cc|c} 2 & 1 & 3 \\ 0 & -4 & 1 \end{array} \right] & \text{(c)} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 5 & 2 \end{array} \right] \\ \\ \text{(d)} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 0 & 1 \end{array} \right] & \text{(e)} \left[\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 5 \end{array} \right] \end{array}$$

- (a) **Yes.** Pivots in columns 1, 2, and 3, stepping right, with zeros below each.
- (b) **Yes.** The pivots are 2 and -4 , not 1, but the definition does not ask for 1s.
- (c) **No.** The all-zero row is not at the bottom (rule 1). Swapping rows 2 and 3 fixes it.
- (d) **No.** The 4 sits directly below the pivot of row 1 (rule 3), so row 2's first nonzero entry is not to the right of row 1's either (rule 2). One elimination step, $R_2 \rightarrow R_2 - 4R_1$, would fix it.
- (e) **No.** Row 1's pivot is in column 2, but row 2's is in column 1, to its *left* (rule 2): the staircase steps the wrong way. Swapping rows 1 and 2 gives

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 5 \end{array} \right],$$

which is in row-echelon form.

Gaussian elimination is the procedure that drives any augmented matrix into this form. The question is what we are **allowed to do** to the rows on the way: only moves that leave the solutions untouched.

Exercise 6.2

Which of these augmented matrices are in row-echelon form? For each one that is not, name a single row operation that would fix it (the operations are listed in the next subsection; a swap or “subtract one row from another” is all you need here).

$$\begin{array}{ll} \text{(i)} \left[\begin{array}{ccc|c} 1 & 4 & -2 & 3 \\ 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] & \text{(ii)} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 4 & 2 \\ 0 & 1 & 1 & 5 \end{array} \right] \\ \\ \text{(iii)} \left[\begin{array}{ccc|c} 0 & 0 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 0 & 1 \end{array} \right] & \text{(iv)} \left[\begin{array}{cc|c} 2 & -1 & 5 \\ 0 & 7 & 0 \end{array} \right] \end{array}$$

Solution to Exercise 6.2

- (i) **Yes.** Pivots in columns 1 and 3 (the step skips column 2, which is allowed), zeros below both, and the zero row at the bottom.
- (ii) **No.** Row 2's pivot is the 1 in column 2, and the 1 directly below it in row 3 should be 0. Fix: $R_3 \rightarrow R_3 - R_2$, which turns row 3 into $[0 \ 0 \ -3 \ 3]$.

- (iii) **No.** The pivots sit in columns 3, 2, 1: the staircase runs backwards. Fix: $R_1 \leftrightarrow R_3$, which gives

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 4 \end{array} \right].$$

- (iv) **Yes.** Pivots 2 and 7, with a zero below the first. The 0 to the right of the bar is just a right-hand side of zero; it has nothing to do with the shape.

6.3 The Row Operations

To reach row-echelon form we must change the numbers **without changing the answer**. These are the moves that do that. Each one is a move we already made on equations in Section 4, now applied to the rows of a matrix. We write R_i for “row i ,” and the arrow $\rightarrow$ for “is replaced by.”

The row operations (each one reversible)

1. **Swap** two rows: $R_i \leftrightarrow R_j$.
Reordering the equations cannot change what solves them.
2. **Multiply or divide** a row by a nonzero constant: $R_i \rightarrow cR_i$, $c \neq 0$.
Doubling both sides of an equation gives an equivalent one. Not by zero, or the equation is destroyed.
3. **Add or subtract** one row to or from another: $R_i \rightarrow R_i \pm R_j$.
Adding equals to equals keeps both sides equal.
4. **Any combination** of these, for example $R_i \rightarrow R_i - cR_j$.
Multiply a row, then subtract it from another: the elimination step from before, and the move used most.

So $R_2 \rightarrow R_2 - 2R_1$ reads “replace row 2 by row 2 minus twice row 1.” Only the row on the left of the arrow changes; the row being used, R_1 here, stays as it was.

Each move can be undone, which is why none of them can lose or invent a solution. A swap is undone by swapping back. Multiplying by c is undone by dividing by c , which is exactly why $c = 0$ is forbidden: multiplying $x + y = 3$ by 0 gives $0 = 0$, which every pair satisfies, and there is no way to divide by zero to get the original equation back. Adding R_j is undone by subtracting it again.

Definition: Row equivalent

Systems related by a sequence of row operations are **row equivalent**, and row-equivalent systems have **exactly the same solutions**.

That guarantee is the whole point. We change the numbers, never the answer: only how easy the answer is to see. The notebook U2-1_Systems-3_GaussianElimination applies each operation step by step.

Example 6.4 (Each operation, applied and checked). Take the system $2x + 4y = 10$, $3x - y = 1$, whose solution is $(1, 2)$ (check: $2 + 8 = 10$ and $3 - 2 = 1$). Apply each kind of operation

and watch what happens to the numbers, and to the solution.

- **Swap.** The same two equations, listed in the other order:

$$\left[\begin{array}{cc|c} 2 & 4 & 10 \\ 3 & -1 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 3 & -1 & 1 \\ 2 & 4 & 10 \end{array} \right]$$

- **Scale.** Row 1 becomes $x + 2y = 5$, every term of $2x + 4y = 10$ halved:

$$\left[\begin{array}{cc|c} 2 & 4 & 10 \\ 3 & -1 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 1 \end{array} \right]$$

- **Replace by a combination.** Starting from the scaled matrix, clear the 3:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -7 & -14 \end{array} \right]$$

Entry by entry: $3 - 3(1) = 0$, $-1 - 3(2) = -7$, $1 - 3(5) = -14$.

Now test $(1, 2)$ in every new row that appeared. In $[3 \ -1 \ | \ 1]$: $3 - 2 = 1$. In $[1 \ 2 \ | \ 5]$: $1 + 4 = 5$. In $[0 \ -7 \ | \ -14]$: $0 - 14 = -14$. The numbers changed at every step; the solution never did. The last matrix is in row-echelon form, so it finishes the problem: $-7y = -14$ gives $y = 2$, then $x + 2(2) = 5$ gives $x = 1$. And each step can be run backwards: $R_2 \rightarrow R_2 + 3R_1$ turns $[0 \ -7 \ | \ -14]$ back into $[3 \ -1 \ | \ 1]$.

Non-examples: moves that are not allowed. Each of these changes the solution set, which is exactly what the legal moves never do.

- **Multiplying a row by zero.** $R_1 \rightarrow 0R_1$ turns $[2 \ 4 \ | \ 10]$ into $[0 \ 0 \ | \ 0]$, and the equation $2x + 4y = 10$ is simply gone. What remains, $3x - y = 1$ on its own, has infinitely many solutions, among them $(0, -1)$, which fails the original first equation: $2(0) + 4(-1) = -4$.
- **Adding a number to every entry.** “Add 1 to row 1” would turn $[2 \ 4 \ | \ 10]$ into $[3 \ 5 \ | \ 11]$, the equation $3x + 5y = 11$. The true solution fails it: $3 + 10 = 13$. Rows may be added to rows, never numbers to rows.
- **Replacing a row without keeping it.** $R_2 \rightarrow 2R_1$ throws row 2 away and puts a copy of twice row 1 in its place, and the information in row 2 is lost for good. In every legal replacement the row being replaced keeps a nonzero multiple of itself: $R_i \rightarrow R_i - cR_j$, or even $R_i \rightarrow 3R_i - 2R_j$, but never $R_i \rightarrow 2R_j$.

Watch out: one operation at a time

Each operation must use the rows **as they currently stand**. It is tempting to save time by doing two at once from the same starting matrix, and that can quietly destroy information. Start from $x + y = 3$, $2x + y = 5$ (solution $(2, 1)$), and perform $R_1 \rightarrow R_1 - R_2$ and $R_2 \rightarrow R_2 - R_1$ “simultaneously,” both computed from the original rows:

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array} \right] \longrightarrow \left[\begin{array}{cc|c} -1 & 0 & -2 \\ 1 & 0 & 2 \end{array} \right] \quad (\text{wrong}).$$

Both new rows say $x = 2$, and y has vanished, so this “system” is solved by $(2, t)$ for every

t , false answers such as $(2, 0)$ included. The new row 2 is just -1 times the new row 1: the second equation was never kept. Done one at a time, the second step uses the *new* row 1: $R_2 \rightarrow R_2 - R_1$ gives $[2 - (-1) \quad 1 - 0 \mid 5 - (-2)] = [3 \quad 1 \mid 7]$, and the system $-x = -2$, $3x + y = 7$ still gives the correct $(2, 1)$.

6.4 The Algorithm, Stated Once

Gaussian elimination

Forward elimination

1. Make **row 1, column 1** into a 1: divide the row by it (if it is 0, swap in a lower row first).
2. Make everything **below** it in that column 0: from each lower row, subtract that row's entry times row 1.
3. Move **down one row and right one column**, to the next entry on the diagonal.
4. Do the same there: make it a 1, then clear everything below it.
5. Repeat until you run out of rows. The 1's are the **pivots**.

Back-substitution

Read the rows back as equations, from the bottom up, substituting each solved variable into the row above.

Step 2 is simple precisely because step 1 made the pivot a 1. If a lower row has a 5 under the pivot, subtracting 5 times the pivot row leaves $5 - 5 \times 1 = 0$ in that spot. In the last row there is nothing below to clear, so making its pivot a 1 is the same division you would do during back-substitution anyway; the worked examples below sometimes leave it until then.

Step 3 describes the usual case, where the next pivot sits on the diagonal. If the diagonal entry is 0 and no lower row has a nonzero entry in that column to swap in, the column gets no pivot at all: stay in the same row, move one more column to the right, and continue from there. That is how the second staircase shape in the previous subsection arises, and Section 7 shows what a pivot-free column means for the solutions.

There is no cleverness anywhere: at each step exactly one thing happens next. That is why a computer can follow it, and why NumPy's `np.linalg.solve` can run it on systems far too large to look at.

Watch out: clear a column with its own pivot row

When you clear below the pivot in column 2, use row 2, never row 1. Row 2 already has a 0 in column 1, so subtracting a multiple of it cannot disturb column 1; row 1 has a nonzero entry there and would undo the work. In the 3×3 example of Section 6.6, at the stage

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 5 & 4 & 1 \end{array} \right],$$

killing the 5 with row 1 ($R_3 \rightarrow R_3 - \frac{5}{2}R_1$) makes row 3 $[-\frac{5}{2} \quad 0 \quad -1 \mid -4]$: the 5 is gone, but

column 1 is no longer clear. Using row 2 instead ($R_3 \rightarrow R_3 - 5R_2$) leaves column 1 alone.

The method carries the name of Carl Friedrich Gauss, who used it in the early 1800s, but it is far older: the Chinese text *The Nine Chapters on the Mathematical Art*, written roughly two thousand years ago, solves systems by laying their coefficients out in a grid on a counting board and combining the columns in just this way.

6.5 A Worked 2×2

Solve $x + y = 3$, $2x + y = 5$, the system whose augmented matrix we wrote above.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 1 & 5 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -1 & -1 \end{array} \right] \xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array} \right]$$

The first row is the **pivot row**, and its leading 1 is already a pivot. To clear the 2 underneath the leading 1, subtract twice the pivot row from row 2, entry by entry: $2 - 2(1) = 0$, $1 - 2(1) = -1$, and $5 - 2(3) = -1$. Scaling row 2 by -1 afterwards is cosmetic: it just makes the pivot a 1.

Back-substitution. Translate the rows back into equations. Row 2 says $0x + 1y = 1$, that is, $y = 1$. Row 1 says $x + y = 3$, and with y known, $x = 3 - 1 = 2$. The solution is $(x, y) = (2, 1)$. Check: $2 + 1 = 3$ and $2(2) + 1 = 5$.

Section 2 of the notebook `U2-1_Systems-3_GaussianElimination` works this example with every step printed by the course's `show_aug` display helper.

Three more 2×2 systems, each adding one new wrinkle: a pivot that needs scaling, a pivot that is zero, and fractions.

Example 6.5 (Scale the first row, then clear). Solve $2x + 6y = 14$, $3x - y = 1$. The top-left entry is 2, so step 1 divides row 1 by 2; every entry in it is even, so no fractions appear. Then clear the 3 below the pivot, and finally scale row 2 to make its pivot a 1:

$$\left[\begin{array}{cc|c} 2 & 6 & 14 \\ 3 & -1 & 1 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & 3 & 7 \\ 3 & -1 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 3 & 7 \\ 0 & -10 & -20 \end{array} \right] \\ \xrightarrow{R_2 \rightarrow -\frac{1}{10}R_2} \left[\begin{array}{cc|c} 1 & 3 & 7 \\ 0 & 1 & 2 \end{array} \right]$$

The middle step, entry by entry: $3 - 3(1) = 0$, $-1 - 3(3) = -10$, $1 - 3(7) = -20$. Back-substitute: row 2 says $y = 2$; row 1 says $x + 3y = 7$, so $x = 7 - 6 = 1$. Check in the originals: $2(1) + 6(2) = 14$ and $3(1) - 2 = 1$. The solution is $(1, 2)$.

Example 6.6 (A zero in the pivot position). Solve $3y = 6$, $2x + y = 8$. Written with both variables, the first equation is $0x + 3y = 6$, so the matrix starts with a 0 in the top-left corner, the pivot position. You cannot divide by 0, and no multiple of a row with 0 in column 1 can clear a nonzero entry below it, so step 1 calls for a swap:

$$\left[\begin{array}{cc|c} 0 & 3 & 6 \\ 2 & 1 & 8 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 2 & 1 & 8 \\ 0 & 3 & 6 \end{array} \right]$$

That single swap already produced row-echelon form: pivots 2 and 3, with a 0 below the first. Rather than divide to make the pivots 1, back-substitute directly: row 2 says $3y = 6$, so $y = 2$; row 1 says $2x + y = 8$, so $2x = 6$ and $x = 3$. Check: $3(2) = 6$ and $2(3) + 2 = 8$. Scaling the pivots to 1 is a convenience, not a requirement.

Example 6.7 (When fractions cannot be avoided). Solve $3x + 2y = 7$, $2x + 5y = 12$. Following the algorithm exactly, divide row 1 by 3:

$$\begin{aligned} \left[\begin{array}{cc|c} 3 & 2 & 7 \\ 2 & 5 & 12 \end{array} \right] &\xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{7}{3} \\ 2 & 5 & 12 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{7}{3} \\ 0 & \frac{11}{3} & \frac{22}{3} \end{array} \right] \\ &\xrightarrow{R_2 \rightarrow \frac{3}{11}R_2} \left[\begin{array}{cc|c} 1 & \frac{2}{3} & \frac{7}{3} \\ 0 & 1 & 2 \end{array} \right] \end{aligned}$$

The middle step: $5 - 2 \cdot \frac{2}{3} = \frac{15}{3} - \frac{4}{3} = \frac{11}{3}$, and $12 - 2 \cdot \frac{7}{3} = \frac{36}{3} - \frac{14}{3} = \frac{22}{3}$. Row 2 says $y = 2$; row 1 says $x + \frac{2}{3}(2) = \frac{7}{3}$, so $x = \frac{7}{3} - \frac{4}{3} = 1$. Check: $3 + 4 = 7$ and $2 + 10 = 12$. The fractions were a nuisance, not an error, and they cancelled in the end.

By hand you can dodge them with a combination move that also scales the row being replaced: $R_2 \rightarrow 3R_2 - 2R_1$ (multiply row 2 by 3, a nonzero number, then subtract twice row 1). Starting again from the original matrix,

$$\left[\begin{array}{cc|c} 3 & 2 & 7 \\ 2 & 5 & 12 \end{array} \right] \xrightarrow{R_2 \rightarrow 3R_2 - 2R_1} \left[\begin{array}{cc|c} 3 & 2 & 7 \\ 0 & 11 & 22 \end{array} \right],$$

since $3(2) - 2(3) = 0$, $3(5) - 2(2) = 11$, and $3(12) - 2(7) = 22$. Row 2 gives $11y = 22$, so $y = 2$, and row 1 gives $3x + 4 = 7$, so $x = 1$: the same answer, with whole numbers throughout. A computer does not mind fractions and follows the plain algorithm; people often prefer the whole-number route.

6.6 A Worked 3×3

Solve

$$\begin{cases} 2x + 4y + 4z = 4 \\ x + 3y + z = 4 \\ -x + 3y + 2z = -1 \end{cases}$$

Clearing the first column. Start by scaling row 1 so the first pivot is a 1. That is optional, but it keeps the arithmetic clean: every entry in row 1 is even, so halving it leaves whole numbers.

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & 4 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array} \right]$$

Now use the pivot row to knock out both entries below it. Row 2 has a 1 under the pivot, so

subtract row 1 once; row 3 has a -1 , so add row 1 once.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ -1 & 3 & 2 & -1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 5 & 4 & 1 \end{array} \right] \end{aligned}$$

Entry by entry, the new row 2 is $1 - 1$, $3 - 2$, $1 - 2$, $4 - 2$, and the new row 3 is $-1 + 1$, $3 + 2$, $2 + 2$, $-1 + 2$.

The first column is done: one pivot on top, zeros beneath it. The bottom two rows are now a two-variable system in y and z alone (their x coefficients are zero), and we repeat the same move one column to the right.

Finishing. The next pivot is the 1 in row 2, column 2, and below it sits a 5, so subtract 5 times row 2 from row 3:

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 5 & 4 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 5R_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 2 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 9 & -9 \end{array} \right]$$

The new row 3 is $0 - 5(0) = 0$, $5 - 5(1) = 0$, $4 - 5(-1) = 9$, and $1 - 5(2) = -9$. That is row-echelon form: pivots in positions (row 1, column 1), (row 2, column 2), and (row 3, column 3), with zeros below each.

Back-substitution, bottom row upward:

- Row 3: $9z = -9 \Rightarrow z = -1$.
- Row 2: $y - z = 2 \Rightarrow y - (-1) = 2 \Rightarrow y = 1$.
- Row 1: $x + 2y + 2z = 2 \Rightarrow x + 2 - 2 = 2 \Rightarrow x = 2$.

The solution is $(x, y, z) = (2, 1, -1)$. Check it in the **original** three equations: $4 + 4 - 4 = 4$; $2 + 3 - 1 = 4$; $-2 + 3 - 2 = -1$. Check against the originals, not against your last matrix: that is what catches an arithmetic slip made three steps ago. A slip made along the way is baked into the final matrix, so an answer checked only against that matrix would look fine.

Section 3 of the notebook `U2-1_Systems-3_GaussianElimination` works this example too, and confirms the answer with `np.linalg.solve`.

Three more full walkthroughs follow, with every row operation shown. The first is clean; the second and third each run into a zero where a pivot should be.

Example 6.8 (A clean 3×3). Solve

$$\begin{cases} x + y + 2z = 6 \\ 2x + 3y + 5z = 13 \\ -x + 2y + 4z = 3 \end{cases}$$

Column 1. The pivot is already 1. Below it sit a 2 and a -1 , so subtract twice row 1 from row

2, and add row 1 to row 3:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 2 & 3 & 5 & 13 \\ -1 & 2 & 4 & 3 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 0 & 1 & 1 & 1 \\ -1 & 2 & 4 & 3 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 0 & 1 & 1 & 1 \\ 0 & 3 & 6 & 9 \end{array} \right]$$

New row 2: $2 - 2$, $3 - 2$, $5 - 4$, $13 - 12$. New row 3: $-1 + 1$, $2 + 1$, $4 + 2$, $3 + 6$.

Column 2. The pivot in row 2 is already 1, with a 3 below it:

$$\xrightarrow{R_3 \rightarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 6 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 6 \end{array} \right]$$

since $3 - 3(1) = 0$, $6 - 3(1) = 3$, and $9 - 3(1) = 6$. That is row-echelon form.

Back-substitution. Row 3: $3z = 6$, so $z = 2$. Row 2: $y + z = 1$, so $y = 1 - 2 = -1$. Row 1: $x + y + 2z = 6$, so $x - 1 + 4 = 6$ and $x = 3$.

Check in the originals: $3 - 1 + 4 = 6$; $6 - 3 + 10 = 13$; $-3 - 2 + 8 = 3$. The three planes meet at the single point $(3, -1, 2)$.

Example 6.9 (A zero pivot halfway through). Solve $x + y + z = 4$, $2x + 2y + 3z = 11$, $x + 3y + 2z = 9$. Column 1 goes as usual:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 2 & 3 & 11 \\ 1 & 3 & 2 & 9 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 1 & 3 & 2 & 9 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 2 & 1 & 5 \end{array} \right]$$

Column 1 is clear. But the next pivot position, row 2, column 2, holds a 0: clearing the $2x$ out of row 2 happened to clear its $2y$ as well. A zero cannot be a pivot, so look below it for a nonzero entry in column 2. Row 3 has a 2, so swap:

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 2 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

That is already row-echelon form; the swap did all the remaining work. Back-substitute: row 3 gives $z = 3$; row 2 gives $2y + 3 = 5$, so $y = 1$; row 1 gives $x + 1 + 3 = 4$, so $x = 0$. Check: $0 + 1 + 3 = 4$; $0 + 2 + 9 = 11$; $0 + 3 + 6 = 9$. The solution is $(0, 1, 3)$, and $x = 0$ is a perfectly good value: zero is an answer, not a failure.

Example 6.10 (Starting with a swap, and keeping track of signs). Solve $2y + 4z = 2$, $x - y + 2z = 5$, $3x - 2y + z = 9$. The first equation has no x , so the matrix opens with a 0 in the pivot position; swap in row 2, which has a 1 there. Then clear the 3 in row 3:

$$\left[\begin{array}{ccc|c} 0 & 2 & 4 & 2 \\ 1 & -1 & 2 & 5 \\ 3 & -2 & 1 & 9 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 2 & 4 & 2 \\ 3 & -2 & 1 & 9 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & -5 & -6 \end{array} \right]$$

Row 3 after the second step: $3 - 3(1) = 0$, $-2 - 3(-1) = 1$, $1 - 3(2) = -5$, $9 - 3(5) = -6$. The second entry is where signs go wrong: subtracting $3 \times (-1)$ adds 3. Row 2 has no x to begin with,

so column 1 is already clear. For column 2, halve row 2 to make its pivot 1, then clear the 1 below it:

$$\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & -5 & -6 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 5 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -7 & -7 \end{array} \right]$$

Back-substitute: $-7z = -7$ gives $z = 1$; $y + 2z = 1$ gives $y = -1$; $x - y + 2z = 5$ gives $x + 1 + 2 = 5$, so $x = 2$. Check: $2(-1) + 4(1) = 2$; $2 + 1 + 2 = 5$; $6 + 2 + 1 = 9$. The solution is $(2, -1, 1)$.

Exercise 6.3

Solve the system

$$\begin{cases} x + 2y = 5 \\ 3x + y = 10 \end{cases}$$

by writing the augmented matrix, reaching row-echelon form with row operations, and back-substituting. Write the row operation you used above each arrow.

Then answer: which single row operation would have been needed first if the two equations had been written in the opposite order?

Solution to Exercise 6.3

The top-left entry is already a 1, so go straight to clearing below it. Row 2 has a 3 under the pivot, so subtract three times row 1 from it; then divide the new row 2 by -5 to make its pivot a 1:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 1 & 10 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -5 & -5 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{5}R_2} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 1 \end{array} \right]$$

Entry by entry, the first step gives $3 - 3(1) = 0$, $1 - 3(2) = -5$, and $10 - 3(5) = -5$.

Back-substitution. Row 2 reads $y = 1$. Row 1 reads $x + 2y = 5$, so $x = 5 - 2(1) = 3$. The solution is $(3, 1)$. Check: $3 + 2(1) = 5$ and $3(3) + 1 = 10$.

The follow-up. None. With the rows reversed the matrix starts as

$$\left[\begin{array}{cc|c} 3 & 1 & 10 \\ 1 & 2 & 5 \end{array} \right],$$

whose top-left entry is 3, which is nonzero and therefore a perfectly good pivot. A **swap** is needed only when the pivot position holds a **zero**. You could still swap by choice, to get a leading 1 and easier arithmetic, but the algorithm does not require it. Following the algorithm to the letter, the first move would be to divide row 1 by 3, which brings in fractions such as $\frac{1}{3}$ and $\frac{10}{3}$; avoiding them is exactly why a swap by choice is tempting.

Exercise 6.4

Solve by Gaussian elimination, showing each augmented matrix and naming each row operation:

$$\begin{cases} x + 2y + z = 4 \\ 2x + 3y + 3z = 7 \\ -x + y + 4z = 7 \end{cases}$$

Solution to Exercise 6.4

Column 1. The pivot is 1; clear the 2 and the -1 below it.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 3 & 3 & 7 \\ -1 & 1 & 4 & 7 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -1 & 1 & -1 \\ -1 & 1 & 4 & 7 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -1 & 1 & -1 \\ 0 & 3 & 5 & 11 \end{array} \right]$$

Column 2. Make the pivot 1 by multiplying row 2 by -1 , then clear the 3 below it.

$$\xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 3 & 5 & 11 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 8 & 8 \end{array} \right]$$

The last step: $5 - 3(-1) = 8$ and $11 - 3(1) = 8$. **Back-substitute:** $8z = 8$ gives $z = 1$; $y - z = 1$ gives $y = 2$; $x + 2y + z = 4$ gives $x + 4 + 1 = 4$, so $x = -1$. **Check in the originals:** $-1 + 4 + 1 = 4$; $-2 + 6 + 3 = 7$; $1 + 2 + 4 = 7$. The solution is $(-1, 2, 1)$.

7 The Degenerate Cases

7.1 What the Three Outcomes Look Like After Elimination

“Degenerate” is the mathematician’s word for a case that falls short of the typical one: here, the systems with no solution or with infinitely many. Below are three systems with nearly the same numbers, each needing just one row operation. Watch the **bottom row**.

One solution. The system $x + 2y = 4$, $2x + 3y = 7$:

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 3 & 7 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & -1 & -1 \end{array} \right]$$

The bottom row reads $-y = -1$. There is a pivot in every column, so $y = 1$, and then row 1 gives $x + 2 = 4$, so $x = 2$.

No solution. The system $x + 2y = 4$, $2x + 4y = 9$:

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 4 & 9 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 1 \end{array} \right]$$

The bottom row reads $0x + 0y = 1$, that is, $0 = 1$: **false**, whatever x and y are. No pair can satisfy it, so the system has no solution. Stop; there is nothing to back-substitute.

Infinitely many. The system $x + 2y = 4$, $2x + 4y = 8$:

$$\left[\begin{array}{cc|c} 1 & 2 & 4 \\ 2 & 4 & 8 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 4 \\ 0 & 0 & 0 \end{array} \right]$$

The bottom row reads $0 = 0$: true but **empty**, since it holds for every x and y and so tells us nothing. That leaves one equation, $x + 2y = 4$, for two unknowns.

Geometrically these are the three pictures from Section 3: crossing lines, parallel lines, the same line. Elimination did not fail in the last two cases; it **reported**. Those two systems have identical left sides, and only the right-hand side (9 versus 8) decided which case we were in.

The same thing happens in three variables, where there are no lines to compare by eye and the row check is the only practical tool. Here is a matching pair of systems that again differ in a single right-hand side.

Example 7.1 (A three-variable system with no solution). Solve $x + y + z = 3$, $2x + 3y + z = 5$, $3x + 4y + 2z = 9$.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 3 & 4 & 2 & 9 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 3 & 4 & 2 & 9 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & 0 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 \end{array} \right] \end{aligned}$$

The bottom row reads $0x + 0y + 0z = 1$: false. **No solution**; stop. Where did the contradiction come from? Add the left sides of the first two equations: $(x + y + z) + (2x + 3y + z) = 3x + 4y + 2z$, which is exactly the left side of the third. So the third left side is forced to equal $3 + 5 = 8$, but the system demands 9. No equation is a multiple of another, so no two of these planes are parallel, yet the three have no common point: this is the tent picture from Section 5, where each pair of planes meets in a line and the three lines never meet.

Example 7.2 (Change one number: infinitely many). Now change the last right-hand side from 9 to 8, so that the third equation is exactly the sum of the first two. The same three row operations give

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 3 & 4 & 2 & 8 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 3 & 4 & 2 & 8 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & -1 & -1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

No contradiction this time: the bottom row is $0 = 0$. Columns 1 and 2 have pivots, column 3 does not, so the system has **infinitely many** solutions, and z will be the free variable. The next subsection writes them all down (Example 7.5). Geometrically, the three planes share a common line, like pages meeting at the spine of a book.

Example 7.3 (Two free variables). Solve $x + 2y - z = 4$, $2x + 4y - 2z = 8$, $-x - 2y + z = -4$.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & 4 & -2 & 8 \\ -1 & -2 & 1 & -4 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \\ -1 & -2 & 1 & -4 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

Two rows vanish completely: the second and third equations were just 2 times and -1 times the first. There is only one pivot, in column 1, so **two** variables are free, y and z . Geometrically all three equations describe the **same plane**, and every point of it is a solution. (Example 7.6 writes them out.)

Watch out: three rows that look alike but mean different things

Row	As an equation	Meaning
$[0 \ 1 \mid 0]$	$y = 0$	a perfectly good value: y is zero
$[0 \ 0 \mid 0]$	$0 = 0$	always true, carries no information
$[0 \ 0 \mid 5]$	$0 = 5$	never true: no solution

A 0 to the right of the bar is harmless; what matters is whether the row has a pivot to the left of the bar. And a row of zeros does not by itself mean “infinitely many”: first look for a contradictory row anywhere in the matrix, then count pivots.

Example 7.4 (More equations than unknowns). Counting equations settles nothing; counting pivots does. Here are three equations in only two unknowns: $x + y = 5$, $x - y = 1$, $2x + y = 8$.

$$\begin{aligned}
 \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 1 & -1 & 1 \\ 2 & 1 & 8 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -2 & -4 \\ 2 & 1 & 8 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -2 & -4 \\ 0 & -1 & -2 \end{array} \right] \\
 &\xrightarrow{R_2 \rightarrow -\frac{1}{2}R_2} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 1 & 2 \\ 0 & -1 & -2 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right]
 \end{aligned}$$

No contradiction, and a pivot in both variable columns: **exactly one** solution. Row 2 gives $y = 2$ and row 1 gives $x = 3$. Check all three: $3 + 2 = 5$, $3 - 2 = 1$, $6 + 2 = 8$. These are three lines that happen to pass through one common point. Change the last right-hand side to 9 and the same steps end in $[0 \ 0 \mid 1]$: the third line now misses the crossing point of the other two, and there is no solution. Going the other way, two equations in three unknowns have at most two pivots for three variable columns, so unless the system is inconsistent there is always a free variable: never exactly one solution.

7.2 Reading the Outcome Off the Rows

After forward elimination, look at the shape

- **A row $[0 \ \cdots \ 0 \mid k]$ with $k \neq 0$:** no solution. Stop.
- **Otherwise, a pivot in every variable column:** exactly one solution. Back-substitute.
- **Otherwise, some variable column has no pivot:** infinitely many solutions, one free variable per pivot-free column.

A “variable column” is any column to the left of the bar. The checks go in this order for a reason: a single contradictory row makes the whole system impossible, so it is looked for first, before anything else is read.

Definition: Free variable

A column with **no pivot** is a **free variable**: set it to anything.

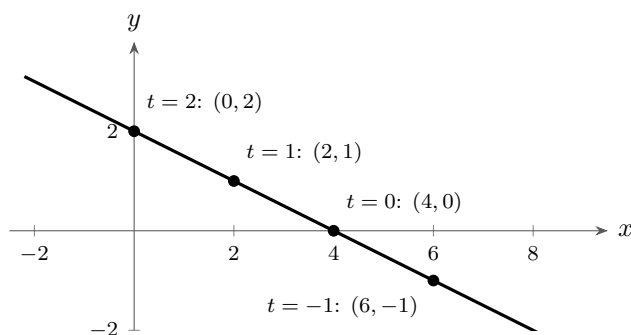
In other words, the variable belonging to a pivot-free column can take any value at all, and the equations then determine the other variables from it. In the third system above, y is free: after

elimination the only pivot is the 1 in row 1, column 1, so column 2 has none. Let $y = t$, where t is a **parameter**, a letter standing for whatever real number you choose. Row 1, $x + 2y = 4$, then gives $x = 4 - 2t$, and every solution is

$$(x, y) = (4 - 2t, t) \quad \text{for every real } t,$$

a parametric description of the line the two equations share. Each choice of t produces one solution: $t = 0$ gives $(4, 0)$, $t = 1$ gives $(2, 1)$, and $t = 2$ gives $(0, 2)$.

Think of the parameter as a dial that slides a point along the line. Every setting of the dial gives one solution, and every solution comes from exactly one setting; no setting gives a point off the line.



The solutions $(4 - 2t, t)$ of $x + 2y = 4$, $2x + 4y = 8$: each value of t picks out one point of the line.

Checking a parametric answer. A parametric solution is checked like any other, except that the check must work for every t at once: substitute the expressions and simplify. For $(4 - 2t, t)$,

$$x + 2y = (4 - 2t) + 2t = 4 \quad \text{and} \quad 2x + 4y = (8 - 4t) + 4t = 8.$$

The t 's cancel in both, so every value of t gives a solution.

This is the same trichotomy as the pictures of lines, but now it is a **mechanical check on the rows** that works in ten variables, where no picture is available.

Example 7.5 (Writing out the solutions of Example 7.2). The reduced matrix was

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right],$$

with no pivot in column 3, so z is free. Let $z = t$. Back-substitute as usual, carrying t along as if it were a known number:

- Row 2: $y - z = -1$, so $y = -1 + t$.
- Row 1: $x + y + z = 3$, so $x = 3 - (-1 + t) - t = 4 - 2t$.

Every solution is

$$(x, y, z) = (4 - 2t, -1 + t, t) \quad \text{for every real } t.$$

Check in the three original equations, for every t at once:

$$\begin{aligned}x + y + z &= (4 - 2t) + (-1 + t) + t = 3, \\2x + 3y + z &= (8 - 4t) + (-3 + 3t) + t = 5, \\3x + 4y + 2z &= (12 - 6t) + (-4 + 4t) + 2t = 8.\end{aligned}$$

The t 's cancel every time. Sample points: $t = 0$ gives $(4, -1, 0)$, $t = 1$ gives $(2, 0, 1)$, and $t = 2$ gives $(0, 1, 2)$. This is the line in space that the three planes share.

Example 7.6 (Two parameters). In the two-free-variable system of the previous subsection, the only surviving row is $x + 2y - z = 4$, and y and z are both free. Give each its own parameter, $y = s$ and $z = t$; then $x = 4 - 2s + t$, and every solution is

$$(x, y, z) = (4 - 2s + t, s, t) \quad \text{for every real } s \text{ and } t.$$

Two independent dials describe a plane, just as one dial described a line. For example, $s = t = 0$ gives $(4, 0, 0)$, and $s = t = 1$ gives $(3, 1, 1)$; check the second in all three original equations: $3 + 2 - 1 = 4$, $6 + 4 - 2 = 8$, $-3 - 2 + 1 = -4$.

Example 7.7 (Any split will do). Back to the shelf with two 5 mg/mL bottles (Example 3.5), now asked for 500 mL at 5 mg/mL:

$$\left[\begin{array}{cc|c} 1 & 1 & 500 \\ 5 & 5 & 2500 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 5R_1} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & 0 & 0 \end{array} \right]$$

Column 2 has no pivot, so y is free: let $y = t$, and row 1 gives $x = 500 - t$. Every solution is $(500 - t, t)$. Here the context adds a restriction the algebra knows nothing about: volumes cannot be negative, so only $0 \leq t \leq 500$ makes sense. Any split of the 500 mL between the two bottles works, which is exactly what the free variable is saying. (With the request for 4 mg/mL instead, the same step gives the row $[0 \ 0 \mid -500]$: no solution.)

Watch out: two traps with free variables

- **“Infinitely many” does not mean “anything goes.”** The system $x + 2y = 4$, $2x + 4y = 8$ has infinitely many solutions, but $(1, 1)$ is not one of them: $1 + 2 = 3 \neq 4$. The solutions fill a line, not the whole plane.
- **Different-looking answers can describe the same set.** Nothing stops you from making x the free variable instead: $x = s$ gives $2y = 4 - s$, so $(x, y) = (s, 2 - \frac{1}{2}s)$. That looks nothing like $(4 - 2t, t)$, but it is the same line: $s = 2$ and $t = 1$ both give $(2, 1)$. The convention in these notes, and in the algorithm, is to take as free exactly the variables whose columns have no pivot, which keeps answers comparable.

Exercise 7.1

What does each augmented matrix say about its system?

$$1. \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array} \right] \quad 2. \left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad 3. \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 7 \end{array} \right]$$

The variables are x, y, z in that order. Where there are infinitely many solutions, describe

them with a parameter.

Solution to Exercise 7.1

For each matrix, first look for a contradictory row $[0 \cdots 0 \mid k]$ with $k \neq 0$, then look for the pivots.

1. **Exactly one solution**, and it is already read off. Row 1 says $1x + 0y + 0z = 4$, that is, $x = 4$; in the same way row 2 says $y = -2$ and row 3 says $z = 3$. There is a pivot in every variable column, and each row has a single variable left in it, so there is nothing to back-substitute: the solution is $(4, -2, 3)$.
2. **Infinitely many solutions**. There is no contradictory row: the bottom row is $0 = 0$, which carries no information. Row 1's pivot is in column 1 and row 2's is in column 3; column 2 has no pivot, so y is **free**. Let $y = t$. Row 2 says $z = 1$, and row 1 says $x + 2y = 5$, that is, $x + 2t = 5$, so $x = 5 - 2t$. The solution set is

$$(x, y, z) = (5 - 2t, t, 1) \quad \text{for every real } t,$$

a line in space.

3. **No solution**. The bottom row says $0x + 0y + 0z = 7$, which is false for every choice of x, y, z . The rows above do not matter: an inconsistent system is inconsistent no matter what else is true.

More practice. Row-reduce each system yourself before reading the solution.

Exercise 7.2

Solve $2x - 4y = 6$, $-3x + 6y = -9$ by Gaussian elimination, and describe every solution.

Solution to Exercise 7.2

$$\left[\begin{array}{cc|c} 2 & -4 & 6 \\ -3 & 6 & -9 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \left[\begin{array}{cc|c} 1 & -2 & 3 \\ -3 & 6 & -9 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + 3R_1} \left[\begin{array}{cc|c} 1 & -2 & 3 \\ 0 & 0 & 0 \end{array} \right]$$

The second step: $-3 + 3(1) = 0$, $6 + 3(-2) = 0$, $-9 + 3(3) = 0$. No contradiction, one pivot, and column 2 is pivot-free, so y is free. Let $y = t$; row 1 says $x - 2y = 3$, so $x = 3 + 2t$. Every solution is $(3 + 2t, t)$. Check for every t : $2(3 + 2t) - 4t = 6$ and $-3(3 + 2t) + 6t = -9$. Both equations were multiples of $x - 2y = 3$ (by 2 and by -3): one line, written twice.

Exercise 7.3

Row-reduce each system and classify it. Where there are infinitely many solutions, write them with a parameter.

1. $x - y + 2z = 1$, $2x + y + z = 5$, $4x - y + 5z = 6$
2. the same, with the last right-hand side 7 instead of 6

Solution to Exercise 7.3

1.

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & 1 & 1 & 5 \\ 4 & -1 & 5 & 6 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 3 & -3 & 3 \\ 4 & -1 & 5 & 6 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 4R_1} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 3 & -3 & 3 \\ 0 & 3 & -3 & 2 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 0 & 3 & -3 & 3 \\ 0 & 0 & 0 & -1 \end{array} \right] \end{aligned}$$

The bottom row says $0 = -1$: **no solution**. The reason: twice the first left side plus the second left side is $(2x - 2y + 4z) + (2x + y + z) = 4x - y + 5z$, the third left side, but $2(1) + 5 = 7$, not 6.

2. With 7 on the right, the same operations give row 3 as $[0 \ 3 \ -3 \ | \ 3]$ after $R_3 \rightarrow R_3 - 4R_1$ (since $7 - 4 = 3$), and then $[0 \ 0 \ 0 \ | \ 0]$ after $R_3 \rightarrow R_3 - R_2$. No contradiction; pivots in columns 1 and 2; z is free. Let $z = t$. Row 2: $3y - 3t = 3$, so $y = 1 + t$. Row 1: $x - (1 + t) + 2t = 1$, so $x = 2 - t$. Every solution is $(2 - t, 1 + t, t)$; for instance $t = 0$ gives $(2, 1, 0)$, which checks: $2 - 1 + 0 = 1$, $4 + 1 + 0 = 5$, $8 - 1 + 0 = 7$.

As in Examples 7.1 and 7.2, identical left sides and a single changed right-hand side flipped the answer from “none” to “infinitely many.”

Exercise 7.4

Solve the system of two equations in three unknowns $x + y + z = 6$, $x - y + 2z = 5$. Then give two specific solutions and check them.

Solution to Exercise 7.4

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \end{array} \right]$$

Pivots in columns 1 and 2; column 3 has none, so z is free. Let $z = t$. Row 2: $-2y + t = -1$, so $y = \frac{1+t}{2}$. Row 1: $x = 6 - y - z = 6 - \frac{1+t}{2} - t = \frac{11-3t}{2}$. Every solution is

$$(x, y, z) = \left(\frac{11-3t}{2}, \frac{1+t}{2}, t \right) \quad \text{for every real } t.$$

Choosing odd values of t gives whole numbers. With $t = 1$: $(4, 1, 1)$, and $4 + 1 + 1 = 6$, $4 - 1 + 2 = 5$. With $t = 3$: $(1, 2, 3)$, and $1 + 2 + 3 = 6$, $1 - 2 + 6 = 5$. Two planes meeting in a line, as Section 5 predicted: never a single point.

8 An Application: Ticket Sales

8.1 Three Kinds of Ticket, Three Facts

“A theater sold 300 tickets: children’s at \$6, adults’ at \$10, seniors’ at \$8, and took in \$2540. As many adult tickets were sold as children’s and seniors’ combined. How many of each?”

There are three unknowns, so name them: c , a , and s are the numbers of children's, adults', and seniors' tickets. Each sentence of the problem is one linear equation:

$$\begin{cases} c + a + s = 300 & \text{(tickets sold)} \\ 6c + 10a + 8s = 2540 & \text{(money taken in)} \\ -c + a - s = 0 & \text{(adults = children + seniors)} \end{cases}$$

The first equation counts tickets. The second counts dollars: c children's tickets at \$6 each bring in $6c$ dollars, and likewise $10a$ and $8s$ for the others. The third fact arrives as $a = c + s$; subtracting c and s from both sides moves everything to the left and puts it in the shape $a_1x_1 + a_2x_2 + a_3x_3 = b$ from Section 2 that the augmented matrix needs, with a 0 on the right. Keep the variables in the same order, c , a , s , in every equation, so that each column of the matrix belongs to one variable.

A count, a price times count, and a comparison: three unknowns need three independent facts, and the problem supplies exactly three.

Most word problems in this course follow the same recipe.

From words to equations

1. **Name the unknowns**, with their units: " c = number of children's tickets." One letter per quantity asked for.
2. **Find one fact per unknown**. The usual kinds are a **total count** ($c + a + s = 300$); a **total amount** built from rates (price times count, concentration times volume, grams per serving times servings); and a **comparison** ("as many as," "twice as many," "20 more than").
3. **Write each fact as an equation**, then move every variable to the left, in the same order, and every constant to the right.
4. **Solve**, then **check the answer against the words**, not just the equations: whole numbers where the context needs them, no negative amounts.

Watch out: comparisons are easy to write backwards

"There are twice as many adult tickets as children's" is $a = 2c$, not $2a = c$. Test a translation with made-up numbers: with 10 children's and 20 adult tickets the sentence is true, and $a = 2c$ agrees ($20 = 2 \times 10$) while $2a = c$ does not ($40 \neq 10$). In standard form, with the variables in the order c , a , the correct translation is $-2c + a = 0$.

8.2 Solving It

Build the augmented matrix and clear the first column with row 1:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 6 & 10 & 8 & 2540 \\ -1 & 1 & -1 & 0 \end{array} \right] & \xrightarrow{R_2 \rightarrow R_2 - 6R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 4 & 2 & 740 \\ -1 & 1 & -1 & 0 \end{array} \right] \\ & \xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 4 & 2 & 740 \\ 0 & 2 & 0 & 300 \end{array} \right] \end{aligned}$$

The new row 2 is $6 - 6$, $10 - 6$, $8 - 6$, $2540 - 1800$, and the new row 3 is $-1 + 1$, $1 + 1$, $-1 + 1$, $0 + 300$.

Either row could be the next pivot row, since both have a nonzero entry in column 2. Swapping puts the smaller numbers on top and keeps everything whole: the row $[0 \ 2 \ 0 \mid 300]$ halves cleanly to $[0 \ 1 \ 0 \mid 150]$.

$$\begin{aligned} & \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 2 & 0 & 300 \\ 0 & 4 & 2 & 740 \end{array} \right] \xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 1 & 0 & 150 \\ 0 & 4 & 2 & 740 \end{array} \right] \\ & \xrightarrow{R_3 \rightarrow R_3 - 4R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 1 & 0 & 150 \\ 0 & 0 & 2 & 140 \end{array} \right] \end{aligned}$$

The last step leaves row 3 as $4 - 4(1) = 0$, $2 - 4(0) = 2$, and $740 - 4(150) = 140$.

Back-substitution, bottom row upward:

- Row 3: $2s = 140$, so $s = 70$.
- Row 2: $a = 150$.
- Row 1: $c + 150 + 70 = 300$, so $c = 80$.

Answer: 80 children's, 150 adults', and 70 seniors' tickets. **Check in the originals:** $80 + 150 + 70 = 300$; $6(80) + 10(150) + 8(70) = 480 + 1500 + 560 = 2540$; $150 = 80 + 70$. All three hold.

Notice that the answers are whole, nonnegative numbers, as ticket counts must be. If a carefully checked calculation had produced a fractional or negative count, that would not be an arithmetic slip to hide: it would say the three facts cannot all be true of a real matinee.

Example 8.1 (What if the facts had not been independent?). Suppose the theater's third fact had instead been "children's and seniors' tickets together numbered 150," so that the system is $c + a + s = 300$, $-c + a - s = 0$, $c + s = 150$, with the revenue left out. Row-reduce, keeping the order c, a, s :

$$\begin{aligned} & \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ -1 & 1 & -1 & 0 \\ 1 & 0 & 1 & 150 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 2 & 0 & 300 \\ 1 & 0 & 1 & 150 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 2 & 0 & 300 \\ 0 & -1 & 0 & -150 \end{array} \right] \\ & \xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 1 & 0 & 150 \\ 0 & -1 & 0 & -150 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 300 \\ 0 & 1 & 0 & 150 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

Row 3 vanished: the new "fact" was already implied by the first two, which between them say $a = 150$ and therefore $c + s = 150$. Column 3 has no pivot, so s is free: with $s = t$, row 2 gives $a = 150$ and row 1 gives $c = 150 - t$. Every $(150 - t, 150, t)$ with t a whole number from 0 to 150 fits all three facts: $(80, 150, 70)$ does, and so does $(100, 150, 50)$. They differ in revenue (\$2540 against \$2500), and that is exactly the information the dependent fact failed to supply. Three equations are not automatically three facts.

8.3 Back to Where We Started

Now the pharmacy question from Section 1: a 10 mg/mL stock and a 2 mg/mL stock, and we need 500 mL at 4 mg/mL. With x mL of the strong stock and y mL of the weak,

$$\begin{cases} x + y = 500 & \text{(volume)} \\ 10x + 2y = 2000 & \text{(milligrams of drug)} \end{cases}$$

It takes one elimination step and one division:

$$\left[\begin{array}{cc|c} 1 & 1 & 500 \\ 10 & 2 & 2000 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 10R_1} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & -8 & -3000 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{8}R_2} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & 1 & 375 \end{array} \right]$$

The first step gives $10 - 10(1) = 0$, $2 - 10(1) = -8$, and $2000 - 10(500) = -3000$; dividing that row by -8 gives 0, 1, and 375.

Row 2 says $y = 375$. Row 1 says $x + y = 500$, so $x = 500 - 375 = 125$. **Draw 125 mL of the strong stock and 375 mL of the weak.**

Check: $125 + 375 = 500$ mL, and $10(125) + 2(375) = 1250 + 750 = 2000$ mg, which is $2000 \div 500 = 4$ mg/mL. It also makes sense: 4 is closer to 2 than to 10, so most of the mix should be the weak stock.

Section 3.1 of the notebook `U2-1_Systems-1_GeometryOfSolutions` solves the same problem and plots it.

Example 8.2 (A request the shelf cannot fill). Suppose the order had asked for 500 mL at 12 mg/mL. The drug total becomes $12 \times 500 = 6000$ mg, and

$$\left[\begin{array}{cc|c} 1 & 1 & 500 \\ 10 & 2 & 6000 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 10R_1} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & -8 & 1000 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{8}R_2} \left[\begin{array}{cc|c} 1 & 1 & 500 \\ 0 & 1 & -125 \end{array} \right]$$

so $y = -125$ and $x = 500 - (-125) = 625$. The algebra is flawless (check: $625 + (-125) = 500$ and $10(625) + 2(-125) = 6250 - 250 = 6000$), and the system has exactly one solution. But that solution calls for drawing -125 mL of the weak stock, which is impossible. The negative number is the math reporting that 12 mg/mL is stronger than the strongest stock on the shelf, and no mixture of a 10 and a 2 can exceed 10. A request is fillable only when its concentration lies between the two stock concentrations, here from 2 to 10 mg/mL.

8.4 More Word Problems

The same recipe handles a wide range of “how much of each” questions. (The clinical numbers in these examples are made up for practice, not dosing guidance.)

Example 8.3 (Mixing dextrose solutions). Dextrose solutions are labeled by percent weight per volume: a 5% solution has 5 g of dextrose in every 100 mL, so each mL carries 0.05 g. A nurse needs 400 mL of 10% dextrose and has 5% and 25% solutions on hand. With x mL of the 5% and y mL of the 25%:

$$\begin{cases} x + y = 400 & \text{(volume, mL)} \\ 0.05x + 0.25y = 40 & \text{(dextrose, g: 10% of 400 mL)} \end{cases}$$

Decimals are legal coefficients, but whole numbers are easier by hand, so first multiply the second equation by 100 (scaling by a nonzero number, a legal move) to get $5x + 25y = 4000$. Then

$$\left[\begin{array}{cc|c} 1 & 1 & 400 \\ 5 & 25 & 4000 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 5R_1} \left[\begin{array}{cc|c} 1 & 1 & 400 \\ 0 & 20 & 2000 \end{array} \right] \xrightarrow{R_2 \rightarrow \frac{1}{20}R_2} \left[\begin{array}{cc|c} 1 & 1 & 400 \\ 0 & 1 & 100 \end{array} \right]$$

Row 2: $y = 100$. Row 1: $x = 400 - 100 = 300$. **Draw 300 mL of the 5% and 100 mL of the 25%.** Check: $300 + 100 = 400$ mL, and $0.05(300) + 0.25(100) = 15 + 25 = 40$ g, which is $\frac{40}{400} = 10\%$. Sensible, too: 10% is much closer to 5% than to 25%, so the mix is mostly the 5%, three parts to one.

Example 8.4 (Tablets that add up to a dose). A regimen calls for 950 mg of a drug per day, taken as exactly 5 tablets, using 250 mg and 100 mg tablets. With a tablets of 250 mg and b tablets of 100 mg, the facts are a count, $a + b = 5$, and a dose, $250a + 100b = 950$:

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 250 & 100 & 950 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 250R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -150 & -300 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{150}R_2} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 1 & 2 \end{array} \right]$$

So $b = 2$ and $a = 5 - 2 = 3$: three 250 mg tablets and two 100 mg tablets. Check: $3 + 2 = 5$ tablets, and $750 + 200 = 950$ mg. Had the answer come out as $a = 2.5$, the regimen would be impossible with whole tablets, and the math would have said so.

Example 8.5 (Enrollment across three clinics). A study enrolled 120 participants at three clinics. Clinic A enrolled twice as many as clinic C, and clinics B and C together enrolled 20 more than clinic A. How many enrolled at each? With a, b, c the three counts, the facts are a total, $a + b + c = 120$; a comparison, $a = 2c$, which becomes $a - 2c = 0$; and another comparison, $b + c = a + 20$, which becomes $-a + b + c = 20$. The middle equation has no b , so its b entry is 0:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 120 \\ 1 & 0 & -2 & 0 \\ -1 & 1 & 1 & 20 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 120 \\ 0 & -1 & -3 & -120 \\ -1 & 1 & 1 & 20 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 120 \\ 0 & -1 & -3 & -120 \\ 0 & 2 & 2 & 140 \end{array} \right] \\ &\xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 120 \\ 0 & 1 & 3 & 120 \\ 0 & 2 & 2 & 140 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 120 \\ 0 & 1 & 3 & 120 \\ 0 & 0 & -4 & -100 \end{array} \right] \end{aligned}$$

The last step: $2 - 2(3) = -4$ and $140 - 2(120) = -100$. Back-substitute: $-4c = -100$ gives $c = 25$; $b + 3c = 120$ gives $b = 45$; $a + b + c = 120$ gives $a = 50$. Check against the words: $50 + 45 + 25 = 120$ participants; clinic A's 50 is twice clinic C's 25; and B and C together, 70, is 20 more than A's 50.

Example 8.6 (Blending tube-feeding formulas). A dietitian blends three enteral formulas into 1000 mL of feed. Per 100 mL, formula A has 4 g of protein and 3 g of fat, formula B has 6 g of protein and 2 g of fat, and formula C has 2 g of protein and 5 g of fat. The blend must deliver 42 g of protein and 31 g of fat. To keep the numbers small, measure each formula in units of 100 mL: let a, b, c be the number of 100 mL portions of A, B, and C, so that 1000 mL is 10 portions. Then

$$\begin{cases} a + b + c = 10 & \text{(volume, in 100 mL portions)} \\ 4a + 6b + 2c = 42 & \text{(protein, g)} \\ 3a + 2b + 5c = 31 & \text{(fat, g)} \end{cases}$$

$$\begin{aligned}
\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 4 & 6 & 2 & 42 \\ 3 & 2 & 5 & 31 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 4R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 0 & 2 & -2 & 2 \\ 3 & 2 & 5 & 31 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 0 & 2 & -2 & 2 \\ 0 & -1 & 2 & 1 \end{array} \right] \\
&\xrightarrow{R_2 \rightarrow \frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & 2 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]
\end{aligned}$$

Back-substitute: $c = 2$; $b - c = 1$ gives $b = 3$; $a + b + c = 10$ gives $a = 5$. Convert back from portions to milliliters: **500 mL of A, 300 mL of B, and 200 mL of C**. Check: $500 + 300 + 200 = 1000$ mL; protein $4(5) + 6(3) + 2(2) = 20 + 18 + 4 = 42$ g; fat $3(5) + 2(3) + 5(2) = 15 + 6 + 10 = 31$ g.

Watch out: convert the units back

Choosing convenient units (here, 100 mL portions) is a good habit, but the answer to the question is in the question's units. Reporting "5, 3, and 2" as milliliters would be off by a factor of 100.

Exercise 8.1

A lab protocol needs 200 mL of a 15% sucrose solution, made from 10% and 30% stock solutions. How much of each stock should be used?

Solution to Exercise 8.1

Let x mL of the 10% and y mL of the 30% be used. Volume: $x + y = 200$. Sucrose: $0.10x + 0.30y = 0.15(200) = 30$ g; multiply by 10 to clear decimals, giving $x + 3y = 300$. Then

$$\left[\begin{array}{cc|c} 1 & 1 & 200 \\ 1 & 3 & 300 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 200 \\ 0 & 2 & 100 \end{array} \right]$$

so $2y = 100$, $y = 50$, and $x = 150$. Use **150 mL of the 10% stock and 50 mL of the 30% stock**. Check: $150 + 50 = 200$ mL, and $0.10(150) + 0.30(50) = 15 + 15 = 30$ g, which is $\frac{30}{200} = 15\%$. The target 15% is a quarter of the way from 10% to 30%, and indeed a quarter of the mix is the strong stock.

Exercise 8.2

A reference lab bills a fixed price for each of three panels: a complete blood count (CBC), a basic metabolic panel (BMP), and a lipid panel. Three invoices read:

- 1 CBC, 1 BMP, and 1 lipid panel: \$75;
- 2 CBCs, 1 BMP, and 3 lipid panels: \$155;
- 3 CBCs, 2 BMPs, and 1 lipid panel: \$140.

Find the price of each panel.

Solution to Exercise 8.2

Let c , b , ℓ be the prices in dollars. Each invoice is one equation: $c + b + \ell = 75$, $2c + b + 3\ell = 155$,

$$3c + 2b + \ell = 140.$$

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 75 \\ 2 & 1 & 3 & 155 \\ 3 & 2 & 1 & 140 \end{array} \right] &\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 75 \\ 0 & -1 & 1 & 5 \\ 3 & 2 & 1 & 140 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 75 \\ 0 & -1 & 1 & 5 \\ 0 & -1 & -2 & -85 \end{array} \right] \\ &\xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 75 \\ 0 & 1 & -1 & -5 \\ 0 & -1 & -2 & -85 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 75 \\ 0 & 1 & -1 & -5 \\ 0 & 0 & -3 & -90 \end{array} \right] \end{aligned}$$

Back-substitute: $-3\ell = -90$ gives $\ell = 30$; $b - \ell = -5$ gives $b = 25$; $c + b + \ell = 75$ gives $c = 20$. **A CBC costs \$20, a BMP \$25, and a lipid panel \$30.** Check all three invoices: $20 + 25 + 30 = 75$; $40 + 25 + 90 = 155$; $60 + 50 + 30 = 140$.

Exercise 8.3

A pharmacy stocks a drug at 20 mg/mL and at 5 mg/mL. An order asks for 300 mL at 25 mg/mL. Set up and solve the system. Can the order be filled? Which concentrations could be filled from these two stocks?

Solution to Exercise 8.3

With x mL of the 20 mg/mL stock and y mL of the 5 mg/mL stock: $x + y = 300$ and $20x + 5y = 25(300) = 7500$.

$$\left[\begin{array}{cc|c} 1 & 1 & 300 \\ 20 & 5 & 7500 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 20R_1} \left[\begin{array}{cc|c} 1 & 1 & 300 \\ 0 & -15 & 1500 \end{array} \right] \xrightarrow{R_2 \rightarrow -\frac{1}{15}R_2} \left[\begin{array}{cc|c} 1 & 1 & 300 \\ 0 & 1 & -100 \end{array} \right]$$

So $y = -100$ and $x = 300 - (-100) = 400$. The system has exactly one solution, and it checks ($400 - 100 = 300$ and $8000 - 500 = 7500$), but it requires -100 mL of the weak stock: **the order cannot be filled.** 25 mg/mL is stronger than the strongest stock, and mixing can only land between the two stock strengths. Any concentration from 5 to 20 mg/mL can be filled; anything outside that range produces a negative volume, exactly as here.

9 Review

What we covered

1. A **linear equation** allows each variable only to the first power, times a constant. In two variables its graph is a **line**; in three, a **plane**.
2. Solving a system means finding the points that satisfy **every** equation, which is where the lines or planes **intersect**.
3. There are exactly **three outcomes**: one solution (crossing), none (parallel, **inconsistent**), or infinitely many (coincident, **dependent**).
4. **Substitution** isolates a variable and feeds it into the other equation. **Elimination** scales and adds equations to make a variable vanish, and is the one that scales up.
5. The **augmented matrix** drops the letters and keeps the numbers, with the right-hand

side after a bar.

6. The **row operations** (swap, multiply or divide, add or subtract, and any combination) produce **row-equivalent** systems, which have identical solutions. That is why elimination is legitimate.
7. **Gaussian elimination** drives the matrix to **row-echelon form** (pivots stepping right, zeros below), then **back-substitution** reads the answer from the bottom row upward.
8. After elimination the outcome is **readable from the rows**: a row $[0 \cdots 0 \mid k]$ with $k \neq 0$ means no solution; a pivot in every variable column means one solution; a pivot-free column means a **free variable** and infinitely many.
9. The method is **mechanical**, which is exactly why `np.linalg.solve` can run it on systems far too big to draw.

Traps to avoid

- Substitute into the **other** equation, and keep the parentheses.
- When scaling or subtracting equations or rows, treat **every** term, right-hand side and signs included.
- A missing variable is a 0 in its column, and every row lists the variables in the same order.
- Do row operations **one at a time**, each on the rows as they currently stand; clear a column with its own pivot row.
- Check answers against the **original** equations, not the last matrix.
- $0 = 0$ is empty, $0 = k$ is impossible, and a 0 to the right of a pivot is simply an answer of zero.
- In word problems, test each translation with made-up numbers, and read a negative or fractional count as a message about the facts.

Notebooks:

- U2-1_Systems-1_GeometryOfSolutions
- U2-1_Systems-2_SubstitutionAndElimination
- U2-1_Systems-3_GaussianElimination

Next: Unit 2-2, **Vectors**.